中柬海洋地学合作

磅逊湾海岸带地质环境图集

ATLAS OF KAMPONG SOM BAY COASTAL GEO-ENVIRONMENT, CAMBODIA

中国地质调查局 China Geological Survey

柬埔寨矿产资源总局 General Department of Mineral Resources, Cambodia

2018.12

前言 | PREFACE

本图集编图范围为柬埔寨磅逊湾西哈努克省行政区范围和近岸海域，覆盖面积约3500km^2。西哈努克省位于柬埔寨西南沿海磅逊湾入口南岸，濒临泰国湾东南侧，海岸线长约180km，占柬埔寨全国岸线长度的三分之一以上。西哈努克省有柬埔寨最大海港西哈努克港，是柬埔寨唯一的现代化商港和外贸的出入门户，也是柬埔寨最繁忙的海岸港口。然而，目前柬埔寨没有专业的海洋地质调查研究机构，海洋地质调查工作几乎空白，这与西哈努克省正在日益突起的海岸带和海洋开发需求无法适应。

2018年3-4月，中国地质调查局青岛海洋地质研究所与柬埔寨矿产资源总局在柬埔寨磅逊湾海岸带联合开展了海岸带地质调查。调查内容包括：海岸带浅地层地震剖面调查、岸滩剖面监测、岸滩及近海海底表层沉积物取样调查、沉积动力调查等。本图集数据来源于本次磅逊湾海岸地质调查数据。图集内容包括海岸带地质和地形地貌、沉积物特征、沉积动力特征、海岸带土地利用状况、岸滩特征、沉积地层等内容。

本图集编制工作由中国地质调查局青岛海洋地质研究所和柬埔寨矿产资源总局共同编制。本图集仅为磅逊湾海岸带地质调查的初步分析成果，限于时间紧促，如有不足之处，敬请谅解。

目 录 | CONTENTS

China-Cambodia Cooperation Agreement

Cooperation Agreement 1

1. Coastal geology and coastal topography

Coastal geology map 2

Coastal topography map 3

2. Surface sediment characteristics

Sediment grain size parameters

Sediment classification 4

Sand content 5

Silt content 6

Clay content 7

Mean grain size 8

Sorting coefficient 9

Skewness 10

Kurtosis 11

Clay mineral characteristics

Chlorite 12

Illite 13

Kaolinite 14

Smectite 15

3. Sedimentary Hydrodynamics

Surface temperature 16

Bottom temperature 17

Surface salinity 18

Bottom salinity 19

Surface turbidity 20

Bottom turbidity 21

Surface dissolved oxygen 22

Bottom dissolved oxygen 23

Surface density 24

Bottom density 25

The variation of the tide after the construction 26

The variation of the tidal prism after the construction 27

The flow field of the flood and ebb tide 28

The tidal residual current field 29

Harmonic constant of K1、O1、M2 tide 30

The distribution of hydrological parameters along the profile 31

4. Land utilization

Land utilization

Land utilization（2001）····32

Land utilization（2010）····33

Land utilization（2017）····34

5. Beach monitoring

Coastal line classification····35

Beach profile map····36

Kampong Som Bay beach distribution····37

6. Sedimentary Stratigraphy and Coastal Geo-hazards

Isopach of the Holocene sediment····38

Distribution of the geo-hazards features····39

7. Work Album

Signing agreement and cooperation workshop····40

China-Cambodia joint field investigation····41

China-Cambodia joint coastal geological survey at Kampong Som Bay····42

Technical training and personnel exchanges····43

Cooperation Agreement

AGREEMENT ON COOPERATIVE PROJECT OF MARINE GEOSCIENCE AND COASTAL GEOHAZARDS BETWEEN THE CHINA GEOLOGICAL SURVEY, THE MINISTRY OF LAND AND RESOURCES, THE PEOPLE'S REPUBLIC OF CHINA AND THE GENERAL DEPARTMENT OF MINERAL RESOURCES, THE MINISTRY OF MINES AND ENERGY, THE KINGDOM OF CAMBODIA

Under the framework of the Memorandum of Understanding (MOU) on Geoscience cooperation signed by China Geological Survey, Ministry of Land and Resources of the People's Republic of China and the General Department of Mineral Resources, Ministry of Mines and Energy of the Kingdom of Cambodia on June 29, 2016, the China Geological Survey, the Ministry of Land and Resources of the People's Republic of China and the General Department of Mineral Resources, the Ministry of Mines and Energy of the Kingdom of Cambodia (hereinafter referred to as the PARTIES) convince that the cooperation in the field of marine and coastal geology should be mutual interest and benefit to each other.

The PARTIES agree to carry out the cooperative project of "Marine Geoscience and Coastal Geo-hazards" (hereinafter referred to as the PROJECT) and reach the following agreement.

ARTICLE 1 PARTIES OF THE AGREEMENT

The PARTIES agree to communicate through their following contacts:

The China Geological Survey, the Ministry of Land and Resources of the People's Republic of China (hereinafter referred to as the CGS)

Address: 45 Fuwai Street, Xicheng District, Beijing
Postcode: 100037
Tel: 86-10-5858 4620
Fax: 86-10-5858 4621
Email: ldawen@mail.cgs.gov.cn

The General Department of Mineral Resources, the Ministry of Mines and Energy of the Kingdom of Cambodia (hereinafter referred to as the GDMR)

Address: #79 89, Pasteur Street, Phnom Penh
Postcode: 12210
Tel: +855-16- 486 88 8
Fax: +855-23-219-584
Email: siengsotham@gmail.com
dica.mme@gmail.com

ARTICLE 2 PURPOSE AND CONTENTS OF THE COOPERATION

The PARTIES agree with the purpose and the content of their cooperation under this Agreement on the cooperative project of Marine Geoscience and Coastal Geohazards as follows:

2.1 Cooperation Purpose

The main purposes of the PROJECT are:

1) To build scientific and technical capacities by sharing China's experiences and methodology on sea-land geological and geophysical mapping, coastal geohazards monitoring and risk assessment, map compilation et al.;
2) To establish a cooperative network and platform for marine geoscientists of the PARTIES;
3) To carry out comparison researches on coastal geohazards characteristics and mechanism for improving capacity of disasters reduction and prevention;
4) To conduct sea-land compilation of geoscience maps series and deep sea structure researches in Cambodia.

2.2 PROJECT Area

The PROJECT will cover the interested coastal and offshore areas, and key regions for comparison study will be selected according to the research focuses.

2.3 Cooperation Contents

2.3.1 Capacity-building

Principles and methodology of coastal and offshore geological survey, monitoring, data processing and database management will be exchanged through training on field survey, laboratory analysis, data management and map production, and data interpretation.

2.3.2 Coastal environment and geo-hazards mitigation

Coastal monitoring systems and risk assessment on groundwater, coastal erosion, land subsidence, salt water intrusion, wetland degradation will be established. Key coastal regions of the PARTIES will be further selected for case comparison, to enhance awareness of the coastal environmental stress and disaster reduction.

2.3.3 Marine geological mapping

The joint coastal-offshore investigation will be carried out in selected demonstration areas for case studies, as well as demonstration of methods and knowledge transfer. At the same time, sea-land geological-geophysical map sheets, such as gravity anomaly, magnetic anomaly, heat flow, seismic tomography, sediment distribution, geo-tectonic and geo-dynamics will be complied though the cooperation of the PARTIES.

2.3.4 Deep sea structure and seismic detection

To fill the data gaps for deep structure study, permanent and temporary seismic stations will be deployed in the cooperative regions of the PARTIES.

2.3.5 Database establishment

To jointly build National Marine Geosciences Database.

ARTICLE 3 PROJECT DURATION AND SCHEDULE

3.1 PROJECT Duration

The duration of this cooperation project is 4 years, from 2017 to 2020.

3.2 Planning and training stage (The first half of 2017)

The PARTIES shall study the existing data on marine geosciences research in the cooperation areas and design the action planning and training the participants.

The PARTIES shall jointly carry out field geological survey in the collaboration study coastal areas in China.

3.3 Training and Field survey stage (The second half of 2017)

The PARTIES shall continue training the participants from Cambodia on methodology and technology of marine geological survey and coastal geohazards monitoring in China.

The PARTIES shall jointly carry out field geological survey in the collaboration study areas in Cambodia.

3.4 Data collection and map compilation stage (2018)

The PARTIES shall choose coastal geohazards monitoring case study area and continue to carry out joint coastal geological survey.

The PARTIES shall deploy the seismic stations in the cooperative region in Cambodia, and collect geological-geophysical data in the cooperative region.

The PARTIES shall preliminarily compile the geoscience map sheets in the joint mapping areas.

3.5 Map compilation and report writing stage (2019-2020)

The PARTIES shall jointly carry out data processing, geoscience map generation, data modeling and researches. The final reports with data and maps should be submitted to both governments by the end of 2020.

ARTICLE 4 OBLIGATIONS OF BOTH PARTIES

4.1 Obligations of CGS

The CGS represents and undertakes to:

4.1.1 To organize and coordinate the CGS team to take part in the PROJECT work.
4.1.2 To be responsible for project design and planning, sustainable operation of technology and labor on the basis of mutual agreement of both PARTIES.
4.1.3 To analyze all collected samples in China laboratories.
4.1.4 To provide the fund for sample analysis of the PROJECT.
4.1.5 To train the members who will participate in the PROJECT on coastal and offshore geological mapping, coastal monitoring and map compilation.
4.1.6 To provide geological-geophysical database in the mapping area as well as geoscience maps compiling and PROJECT report writing.
4.1.7 In order to implement the PROJECT, Chinese Party shall provide with customs clearance, and taxation for equipments, work utensils and necessary items to be carried by GDMR when entering into China.
4.1.8 To give official entry permission to GDMR team to China for PROJECT work.

4.2 Obligations of GDMR

The GDMR represents and undertakes to:

4.2.1 To organize and coordinate the GDMR team to take part in the PROJECT work.
4.2.2 To provide CGS team with PROJECT information as needed, such as geological maps, topographic maps, distribution of licenses, geological-geophysical data, as per the needs of this PROJECT.
4.2.3 To assist CGS team in obtaining entry visa to Cambodia for PROJECT work.
4.2.4 To coordinate with the governmental authorities and local community as well as local people to ensure smooth PROJECT implementation in Cambodia.
4.2.5 To assist CGS to deliver or ship the necessary materials, maps, and samples related to the PROJECT to China, subject to existing and applicable Cambodian laws, rules and regulations.
4.2.6 To relief tax of equipment sent to Cambodia for completing the PROJECT by the CGS team as a share of the Cambodian government for the PROJECT.
4.2.7 To provide other necessary helps under the authority of GDMR during the implementation of the PROJECT.

4.3 Mutual Obligations

4.3.1 Both PARTIES shall collaboratively set up detailed action plan for each stage, and undertake field investigations, field research and general study as well as sending relevant specialists and providing necessary information.
4.3.2 Both PARTIES shall collaboratively implement the training course on the Principle and Methodology of coastal-offshore geological-geophysical Mapping, coastal monitoring, map compilation to PROJECT participants of the PaARTIES.
4.3.3 Both PARTIES shall collaboratively to give data interpretation and target delineation.
4.3.4 Both PARTIES shall convene non-periodic project meetings as needed by the PROJECT development, and discuss collaborative problems existing in the PROJECT development and investigate the following work arrangement.

ARTICLE 5 FINANCIAL ARRANGEMENTS

5.1 Both PARTIES shall provide the necessary salary, bonus and other emoluments for their own personnel involved in the PROJECT.
5.2 During the joint research and field work in Cambodia, CGS shall provide budget for accommodation, insurance and international flights and field work cost for CGS staff, and GDMR shall provide budget for accommodation and daily allowance for GDMR participants.
5.3 During the joint research, field work and technical training in China, CGS shall provide the funding support if necessary according to the annual working action for Cambodian participants.

ARTICLE 6 ACHIEVEMENTS

The product of the project includes:

1. The Regional-scale Geological-geophysical Data and a series of sea-land maps and the instruction;
2. The Thematic Report of Case Study in coastal regions;
3. The Final Marine Geoscientific Report of the PROJECT.

ARTICLE 7 WORKING LANGUAGE

The working language of the PROJECT is English. Training, meeting, and resolutions and summaries formed in the meetings, as well as mutually exchanged information and achievements, etc., shall be expressed in English.

ARTICLE 8 SHARING AND PROTECTION

The protection and distribution rules of data obtained from the cooperation under this agreement shall abide by relevant Intellectual Property laws of China and Cambodia and relevant international convention or bilateral treaty.

ARTICLE 9 DUTIES OF NON-COMMITMENT AND TERMINATION OF THE AGREEMENT

9.1 Both PARTIES shall be free from any responsibilities concerning failure or non-performance due to force majeure (such as war, turmoil, earthquake, devastating disaster or any other natural disaster…etc) which is out of the government's control or the factors which are out of both parties' control.
9.2 Termination date of the PROJECT refers to finish time of the work or specified date in the agreement, whichever comes first. However, under the circumstances which are agreed upon by both PARTIES, the publication of the work achievement reports may be extended beyond the expiration of this agreement.

ARTICLE 10 PROJECT COORDINATION

10.1 The PARTIES shall establish a working group, formulate a road map of the PROJECT, and discuss specific cooperation details.
10.2 The PARTIES shall respectively designate one person as PROJECT coordinator, who shall be in charge of daily contact, negotiating each other on behalf of his country, exchanging information and documents, and organizing domestic experts to finish the tasks which are undertaken by each country as per the confirmed plan.

10.2.1 The Chinese coordinator
Name: Liu Dawen
China Geological Survey
Address: 45 Fuwai Street, Xicheng District, Beijing
Postcode: 100037
Tel: 86-10-5858 4660
Fax: 86-10-5858 4661
Email: ldawen@mail.cgs.gov.cn

10.2.2 The GDMR Coordinator
Name: Mr. Sieng Sotham
Address: #79-89, Pasteur Street, Phnom Penh
Postcode:12210
Tel: +855-16- 486 888
Fax: +855-23-219-584
Email: siengsotham@gmail.com
dica.mme@gmail.com

ARTICLE 11 DISPUTE RESOLUTION

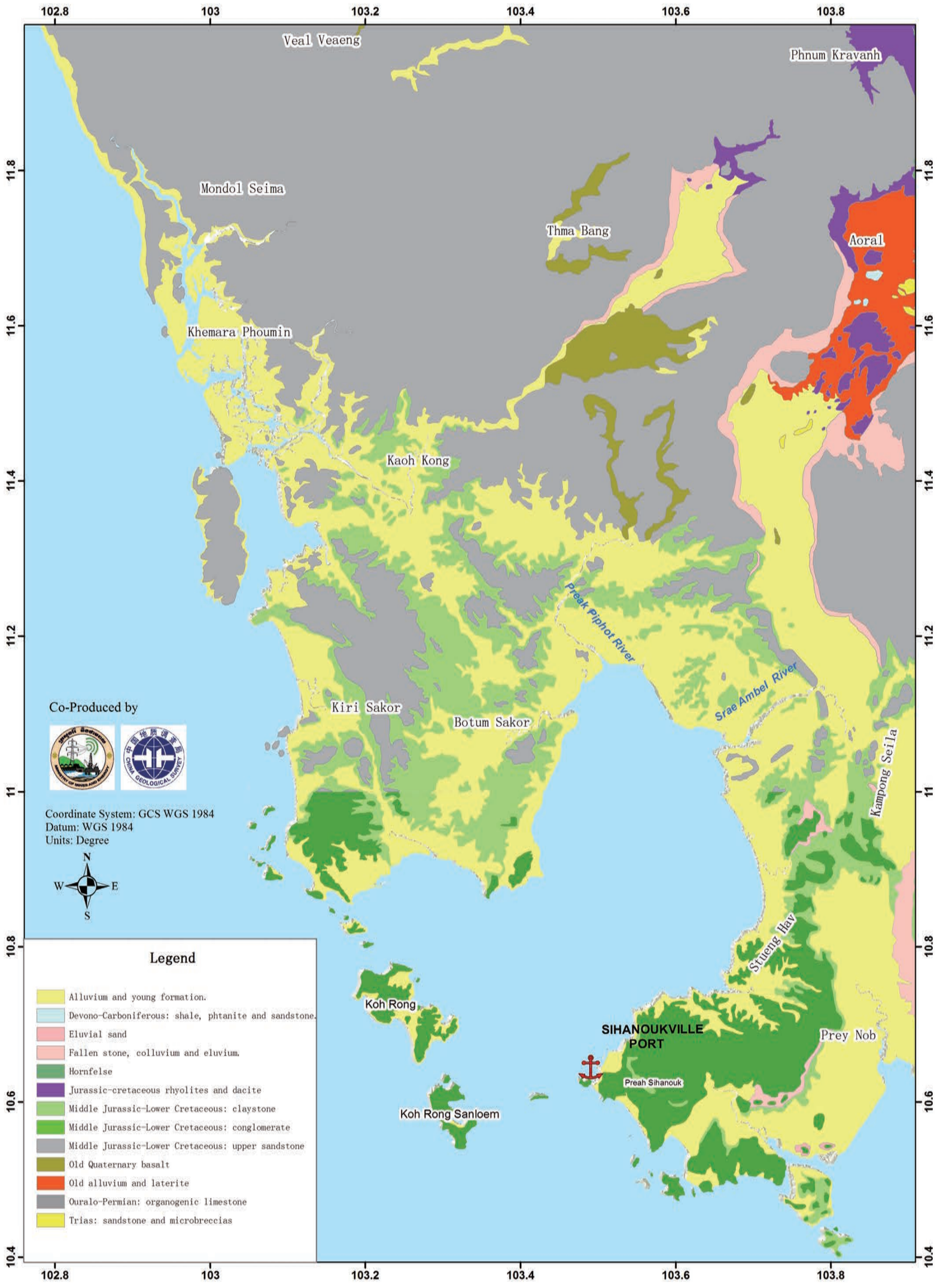
Any dispute arising from the performance of duties committed by both PARTIES in this agreement shall be firstly settled through friendly negotiation.

ARTICLE 12 MISCELLANEOUS

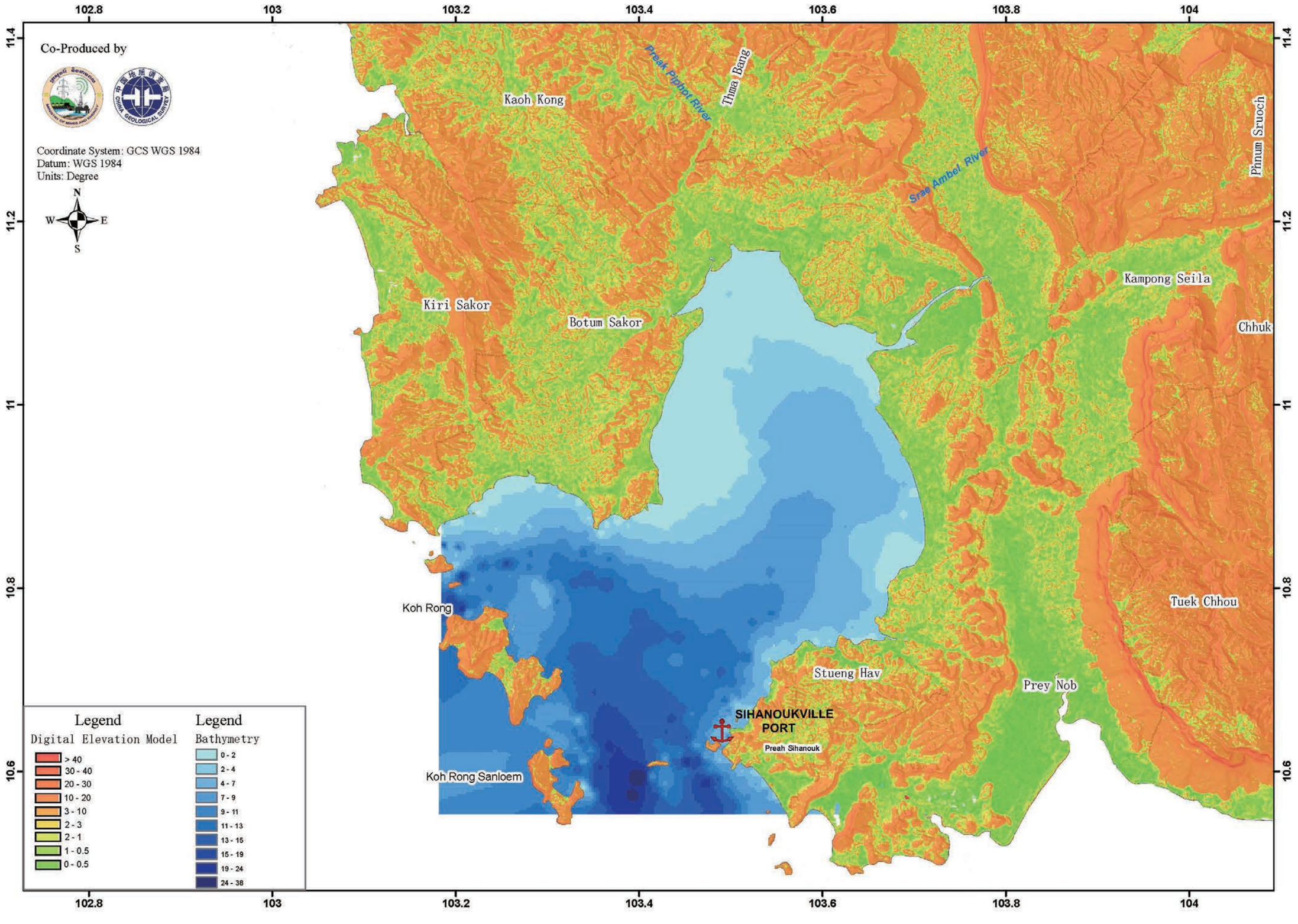
12.1 Resolutions and summaries formed in the work meetings and coordination conferences collaboratively convened by both PARTIES are significant supplements to this agreement and are equally binding to both PARTIES.
12.2 This agreement is signed in English on date of April 18, 2017 in Beijing. Each PARTY holds one copy of the agreement.
12.3 This agreement becomes effective since the date on which representatives of cooperation PARTIES sign and approve this agreement.

For and on behalf of China Geological Survey Ministry of Land and Resources People's Republic of China Duly authorized representative	For and on behalf of General Department of Mineral Resources Ministry of Mines and Energy Kingdom of Cambodia Duly authorized representative
Name: Li Jinfa Position: Vice-president	Name: UNG Dipola Position: Deputy Director General

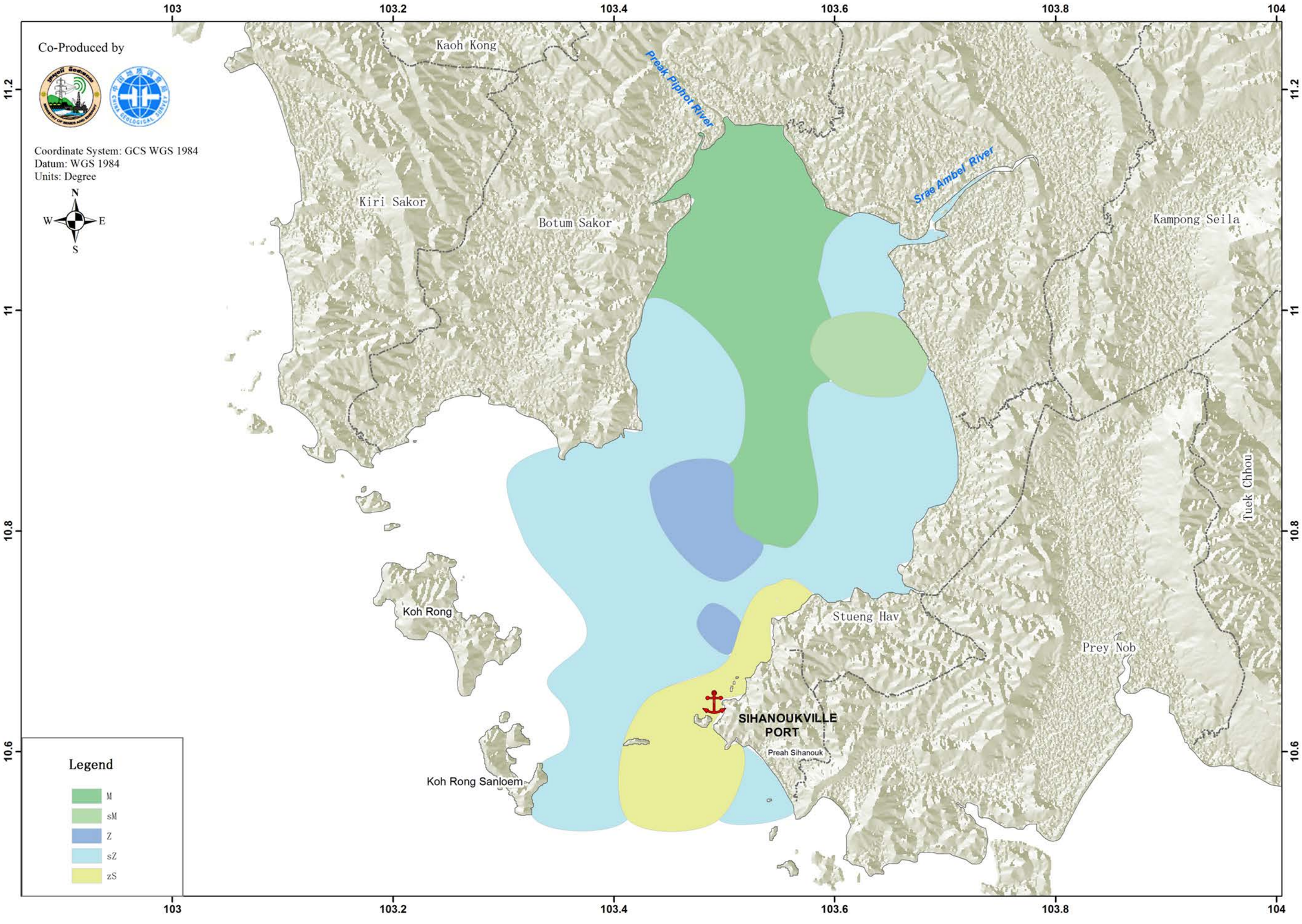
Coastal Geology Map



Coastal Topography Map



Sediment Classification



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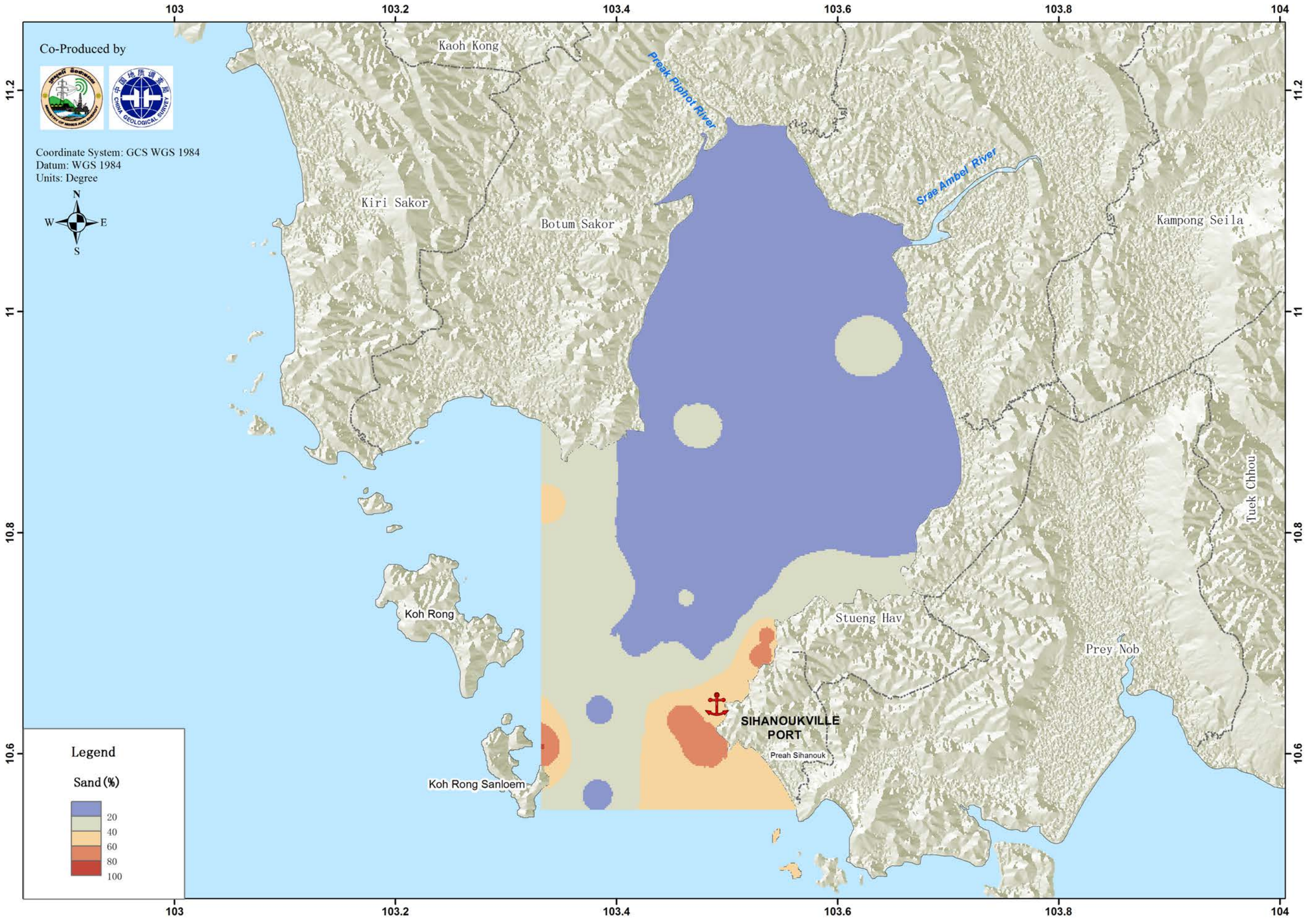

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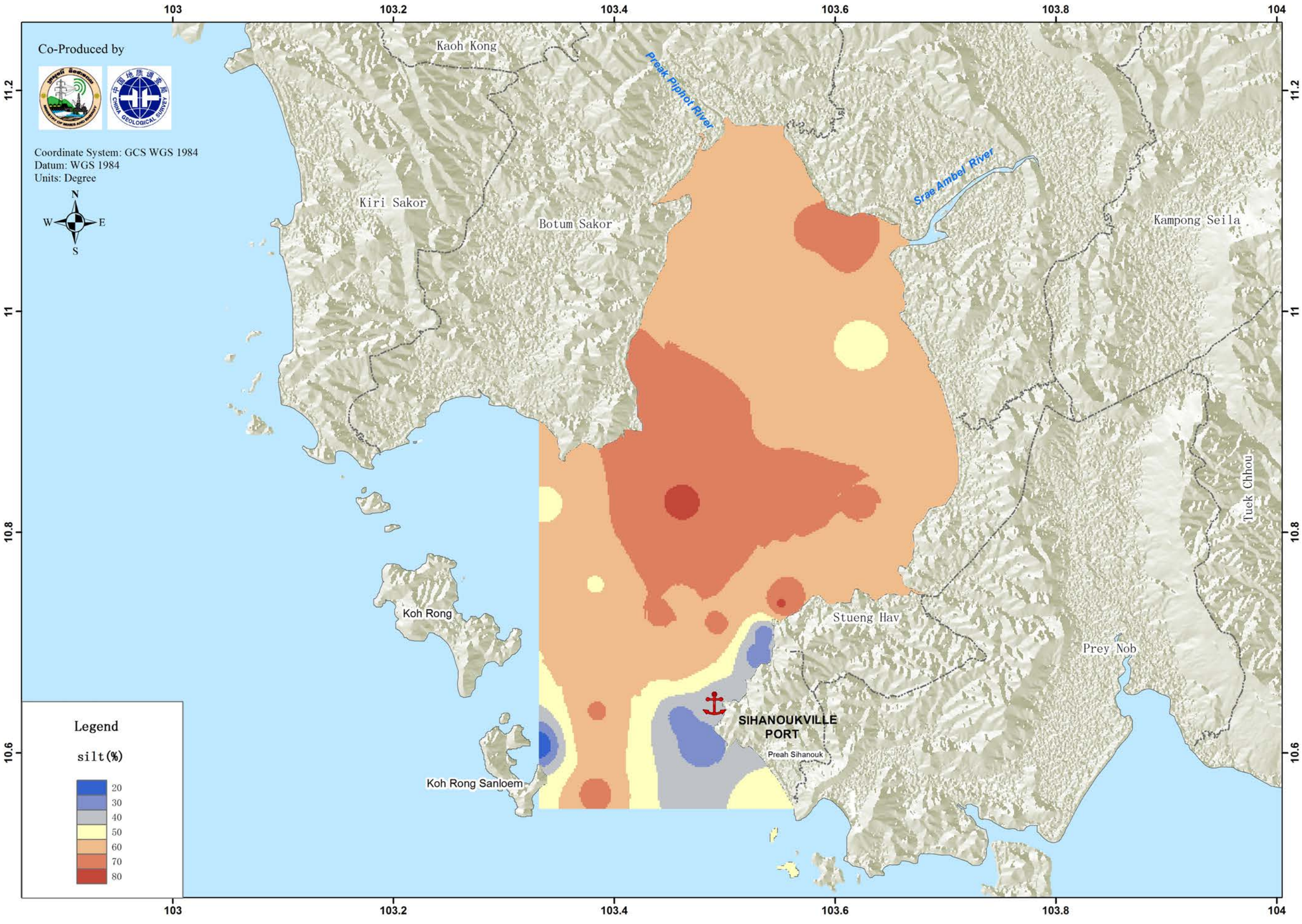

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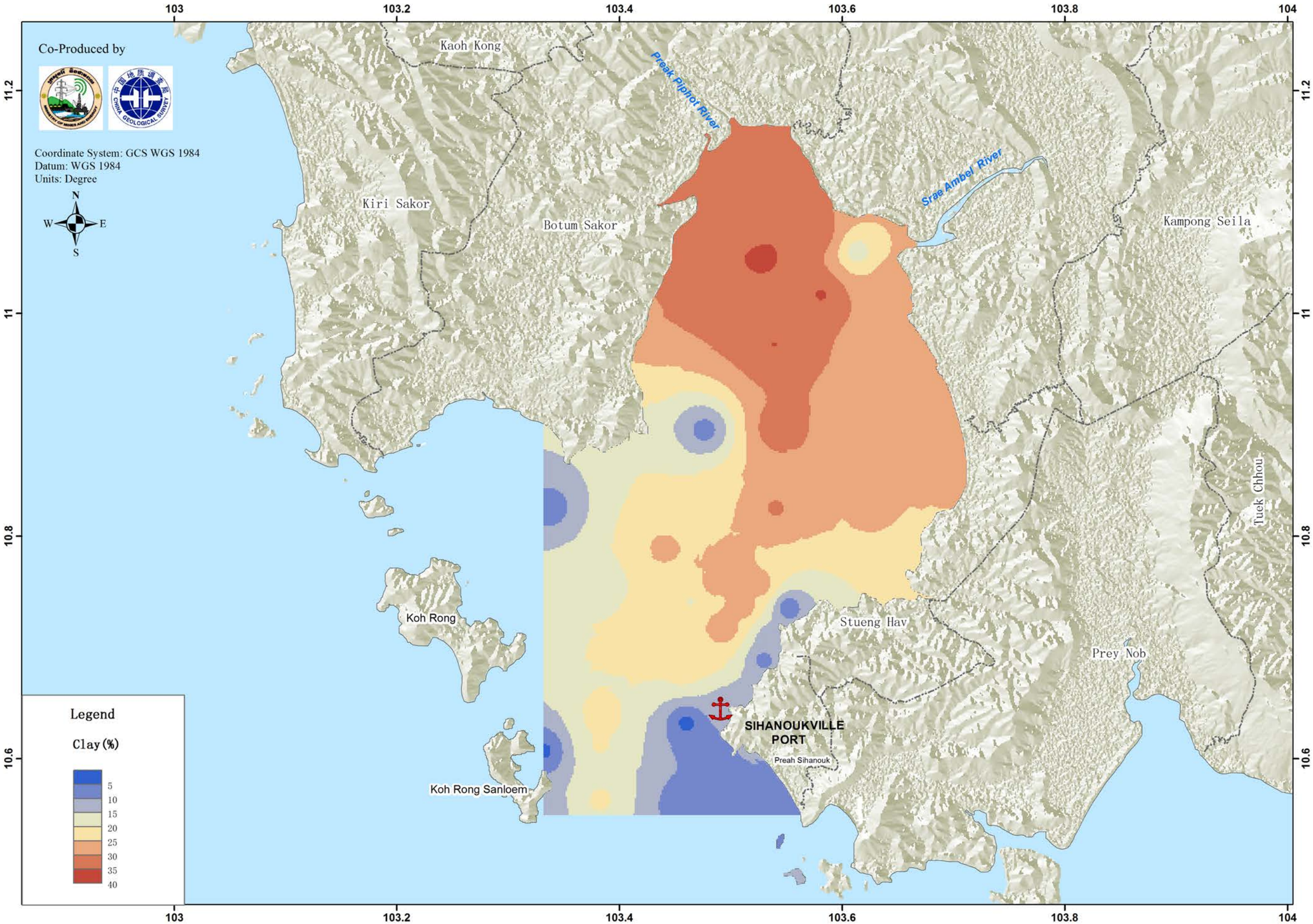
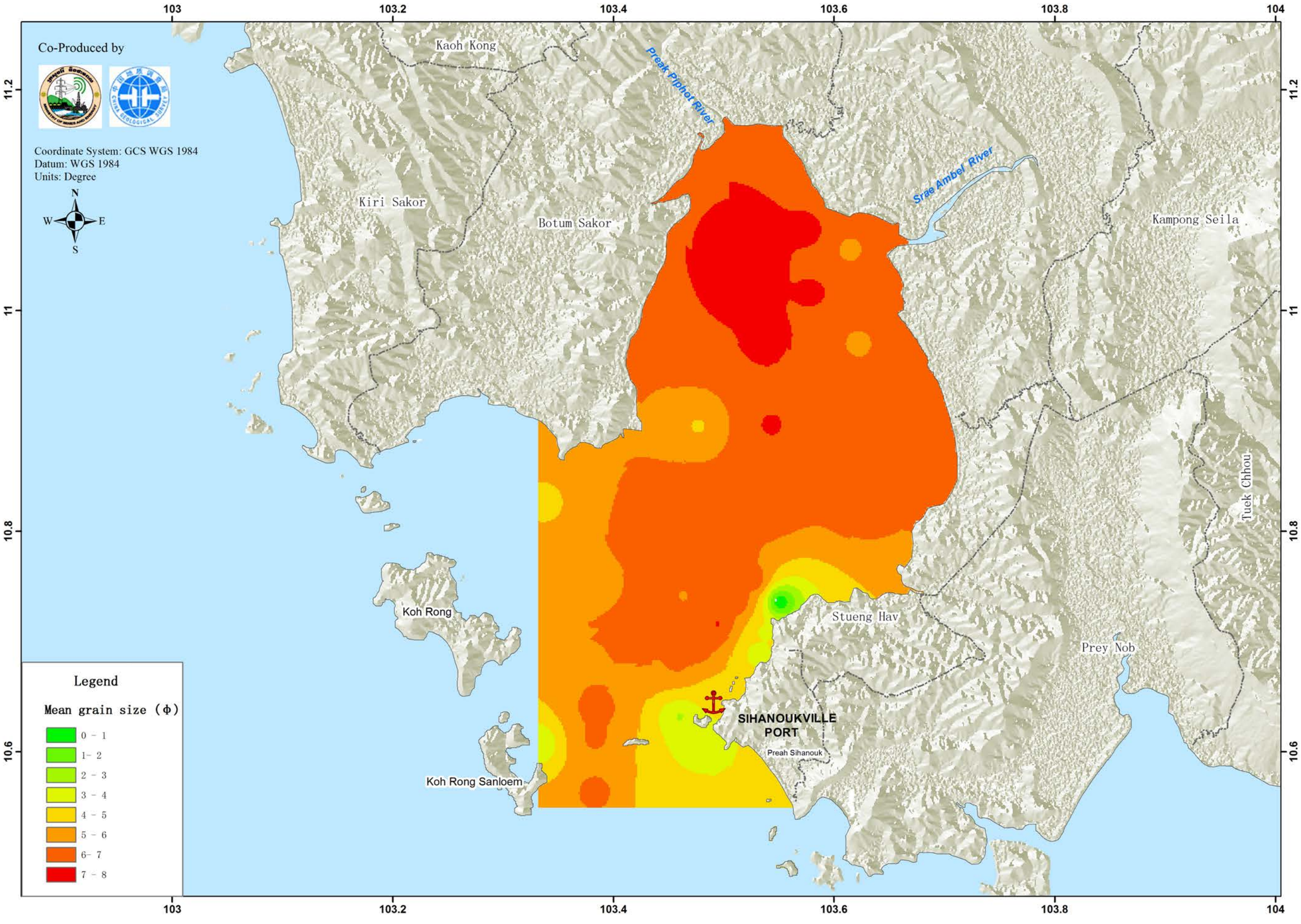
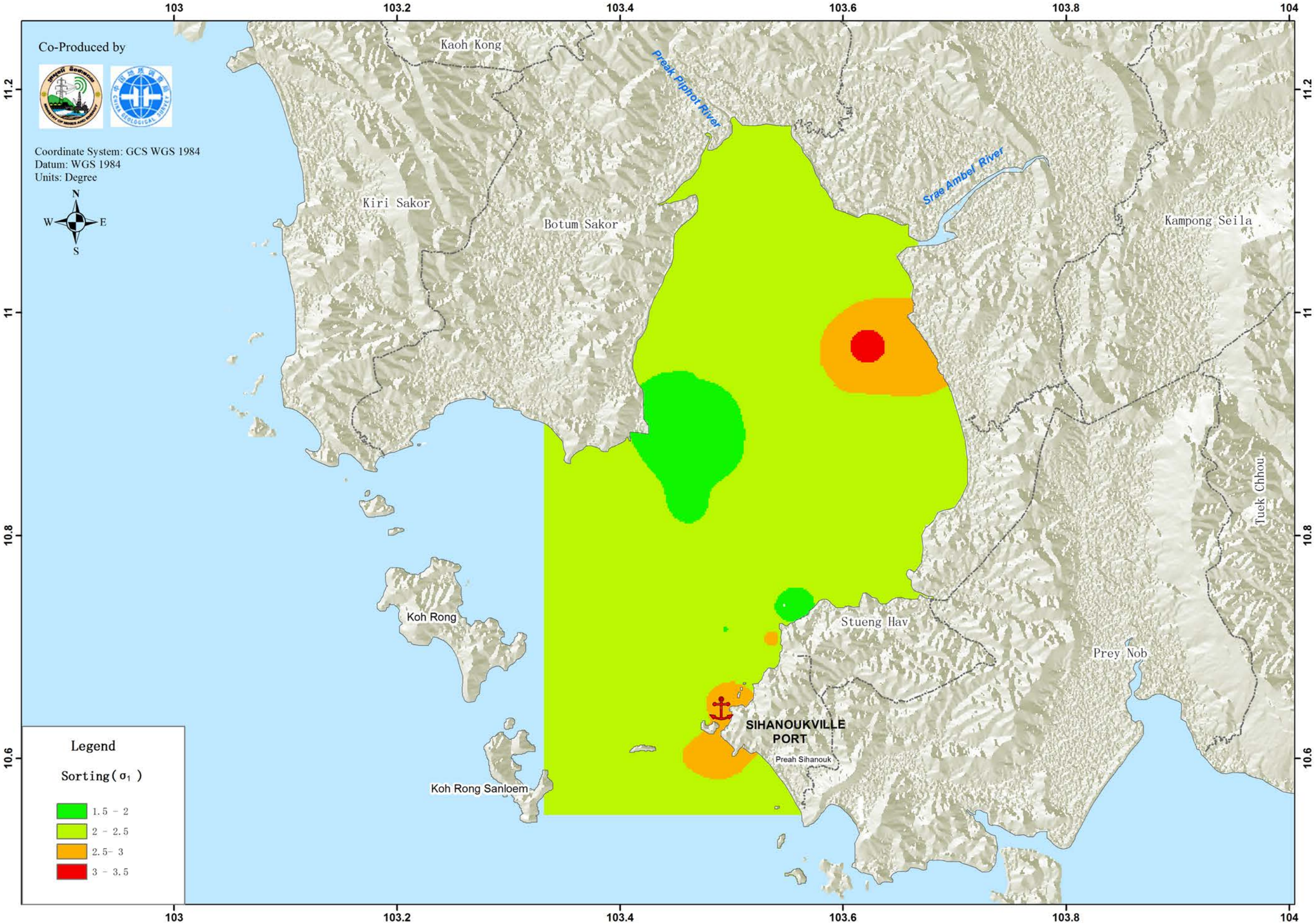


Mean Grain Size



Sorting Coefficient



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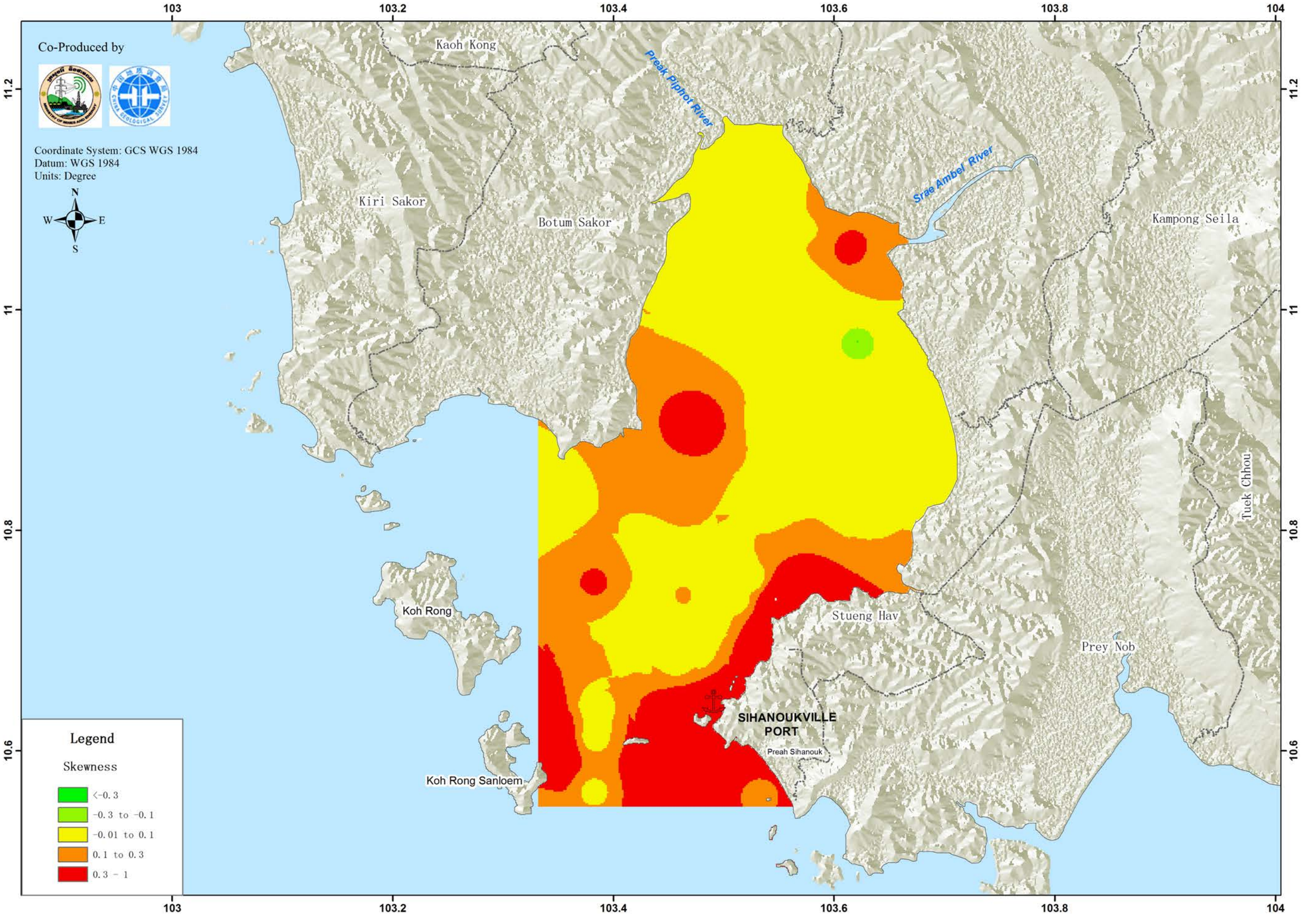
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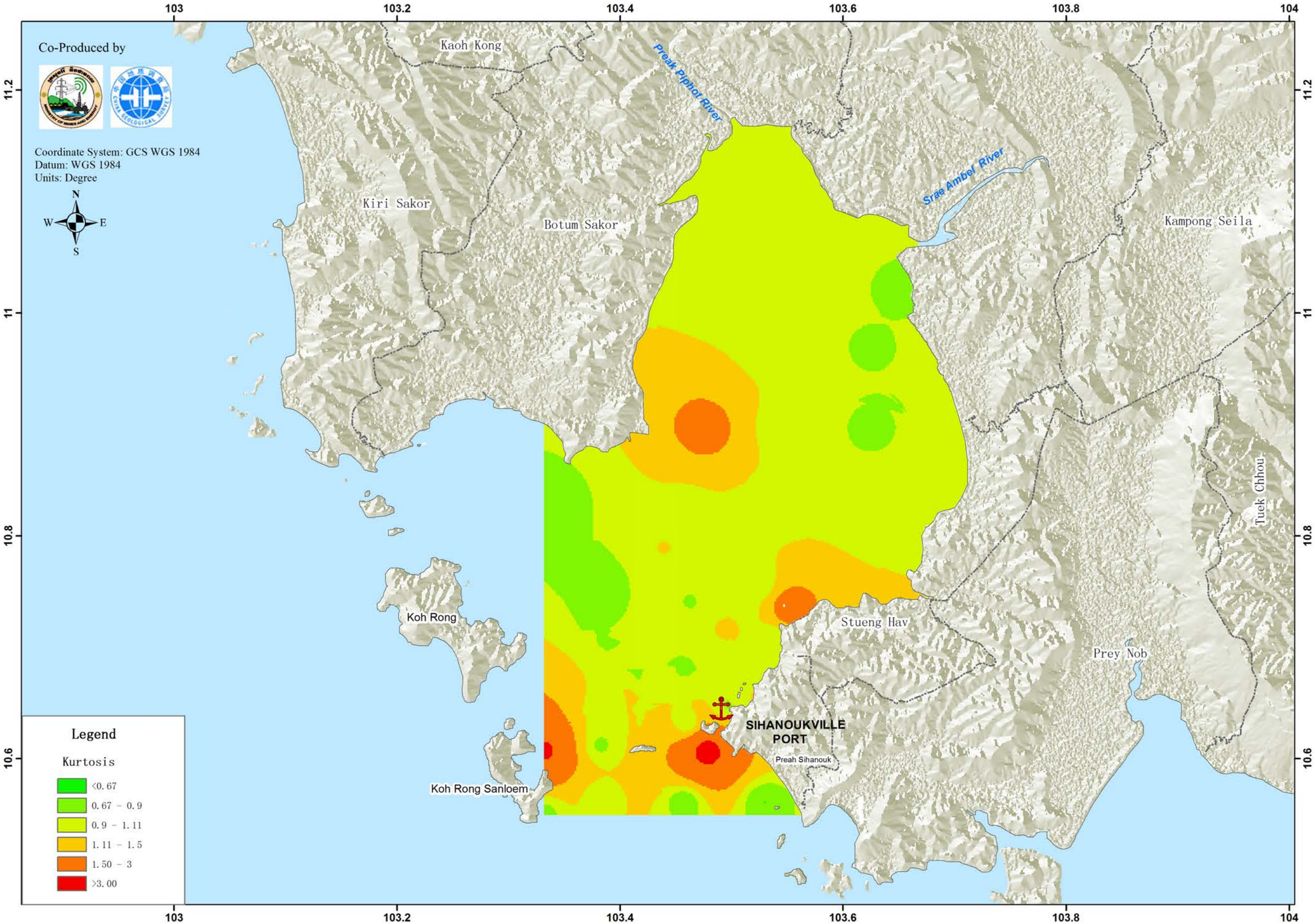
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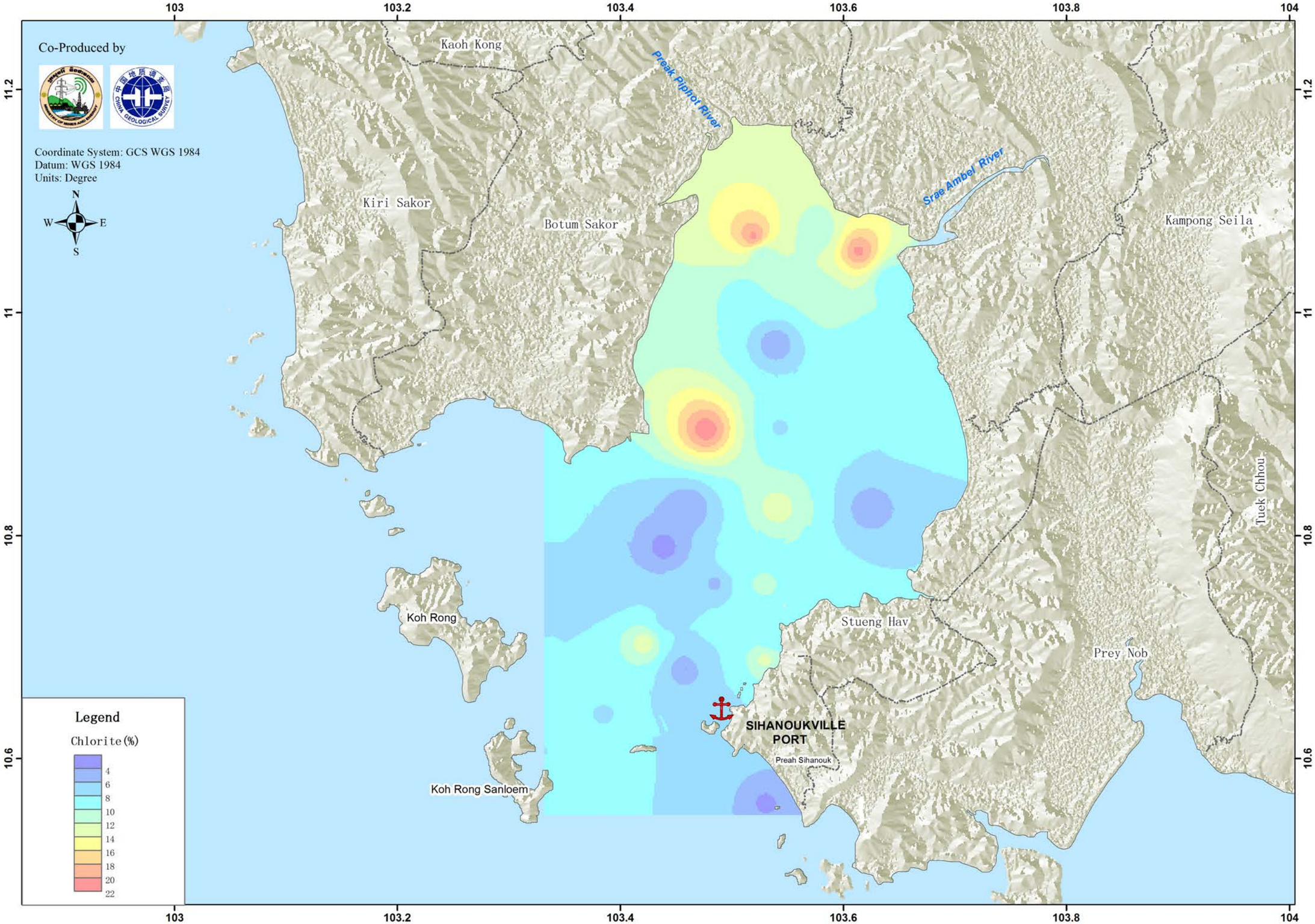
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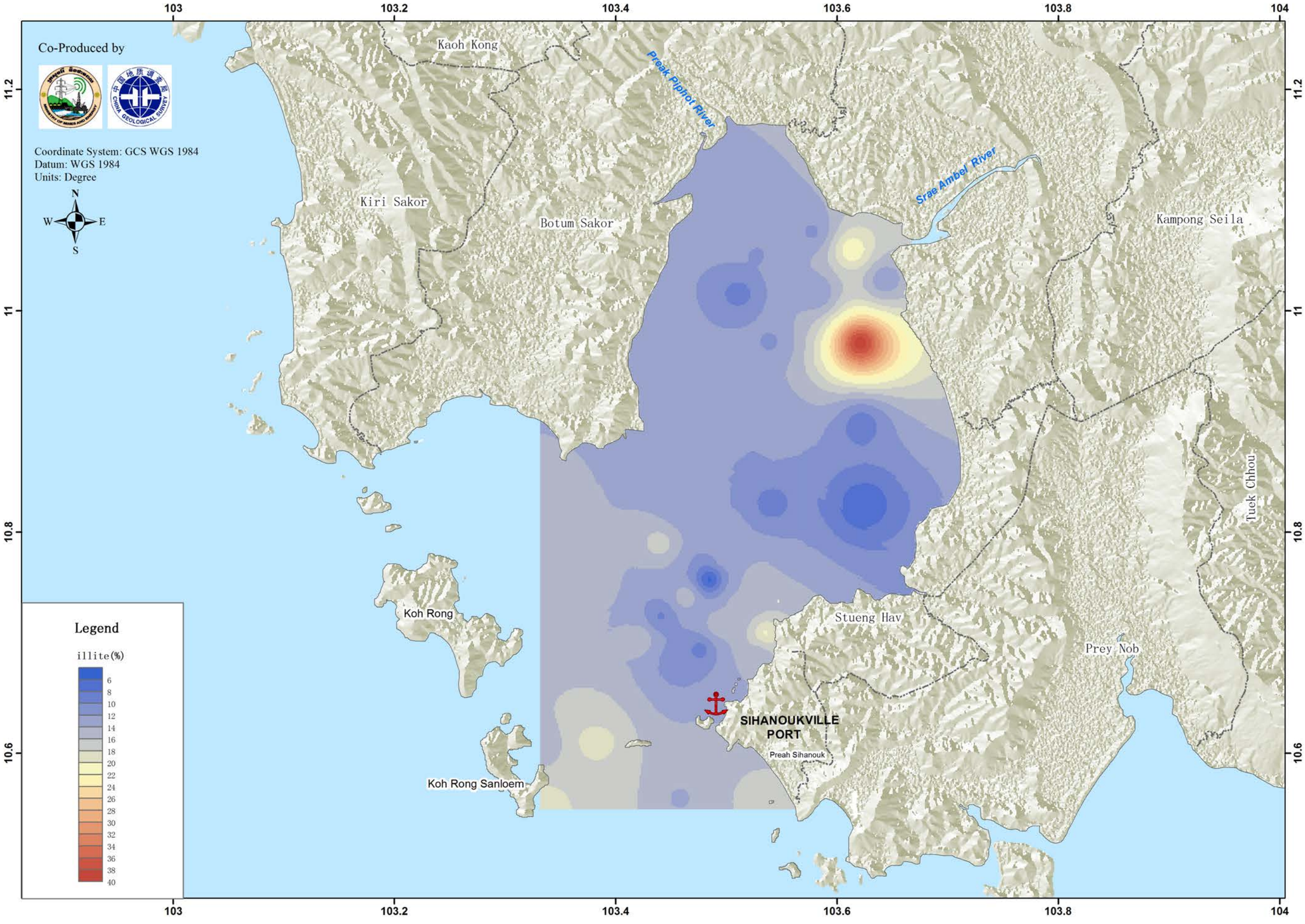
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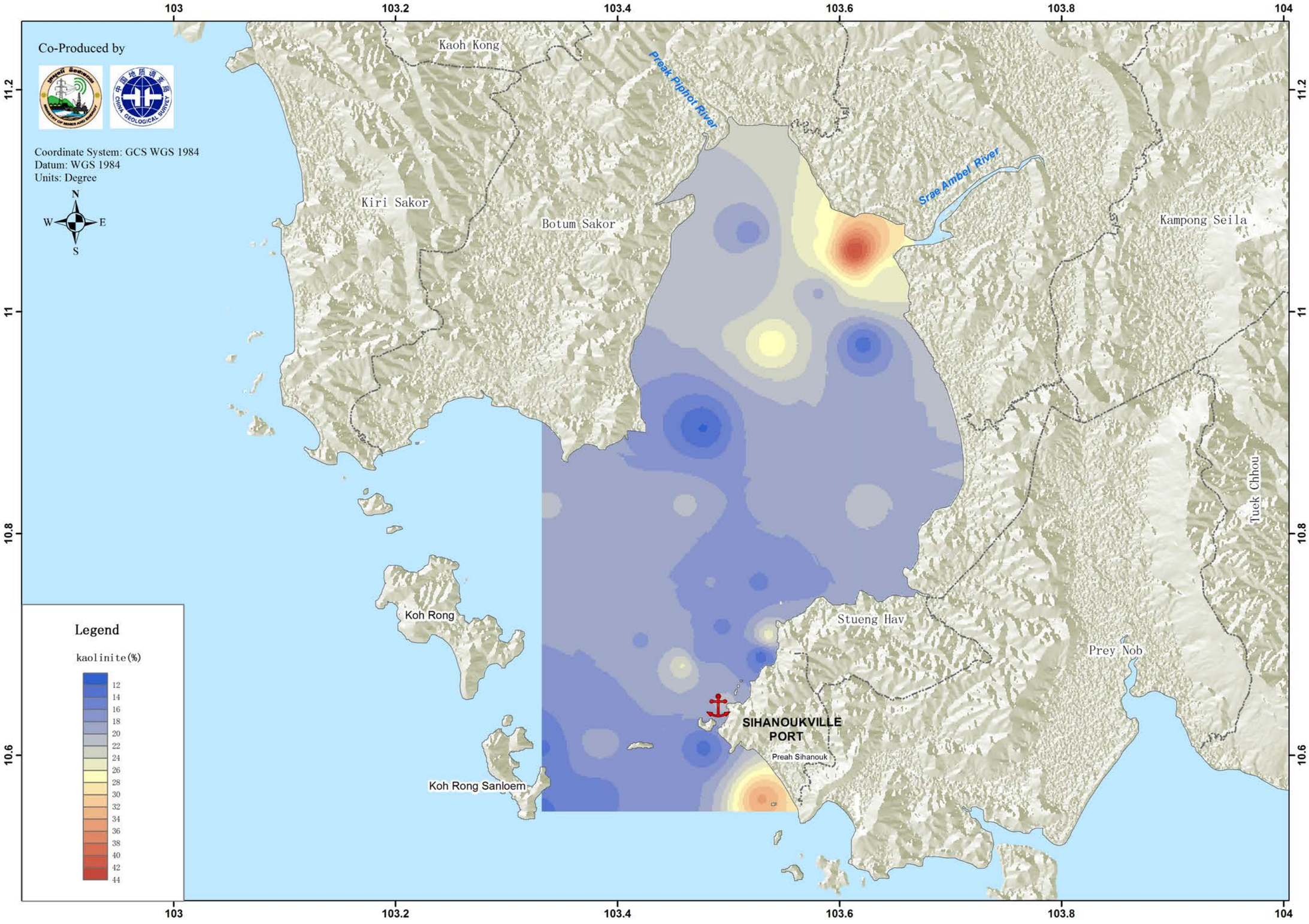
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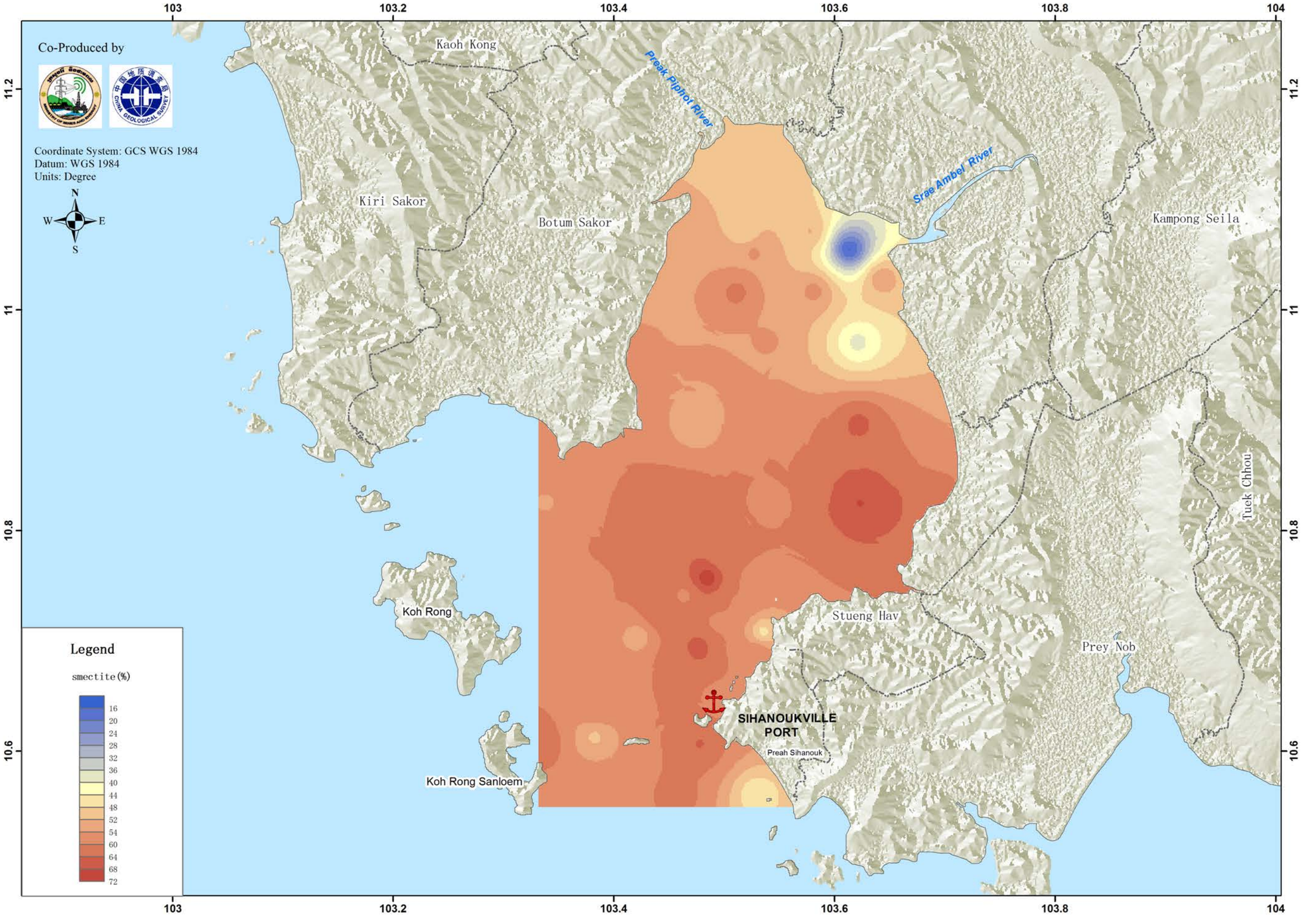
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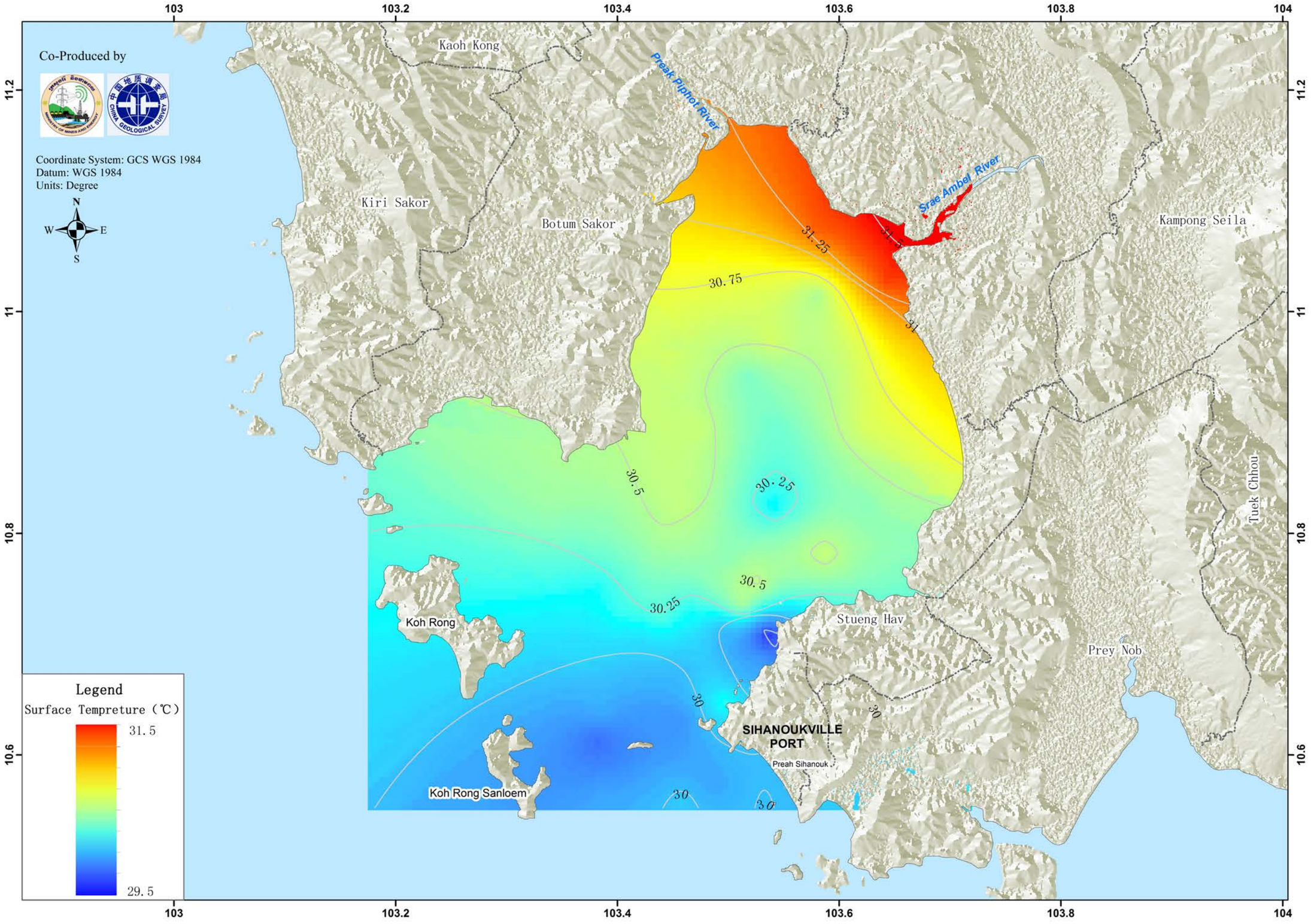
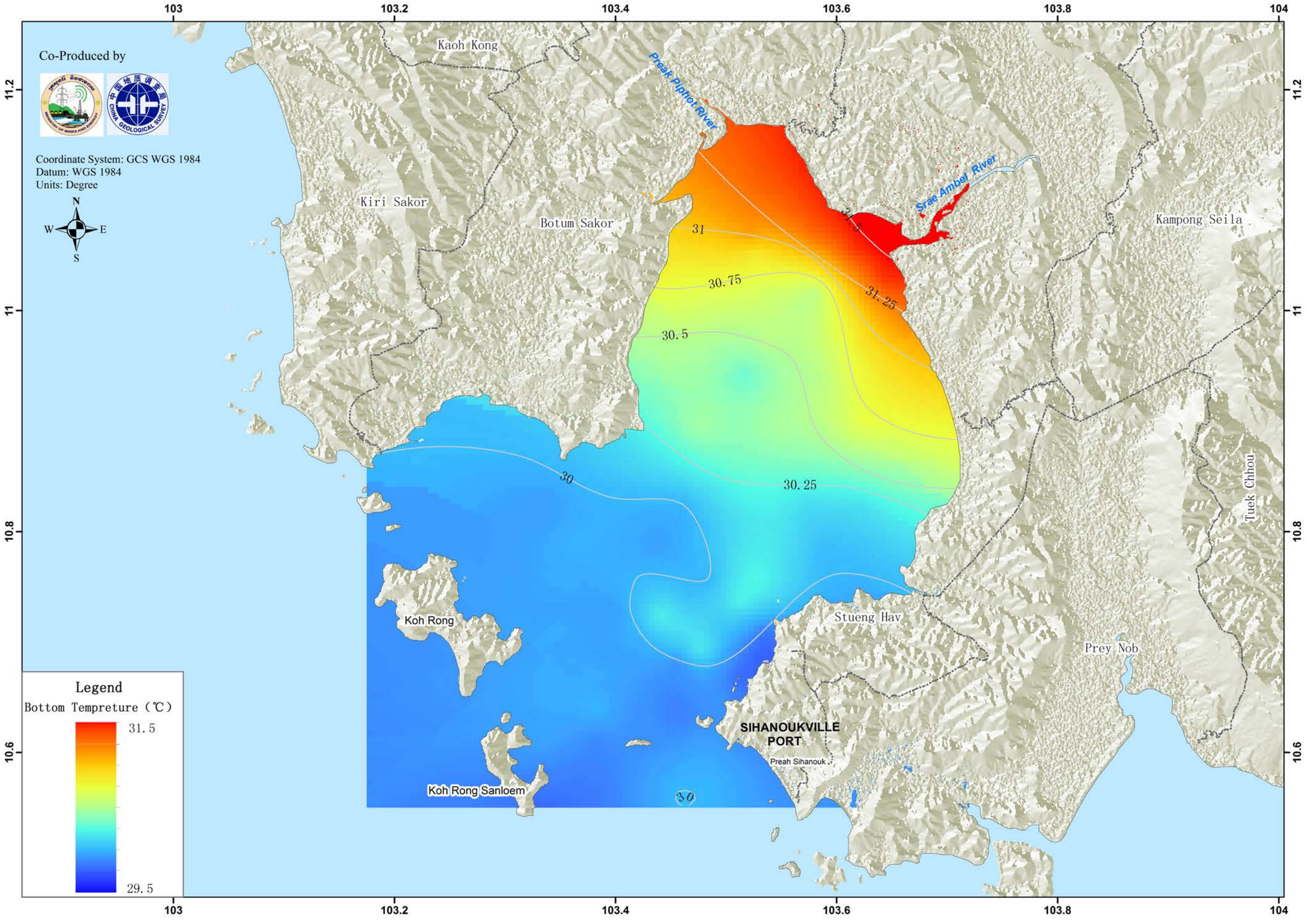
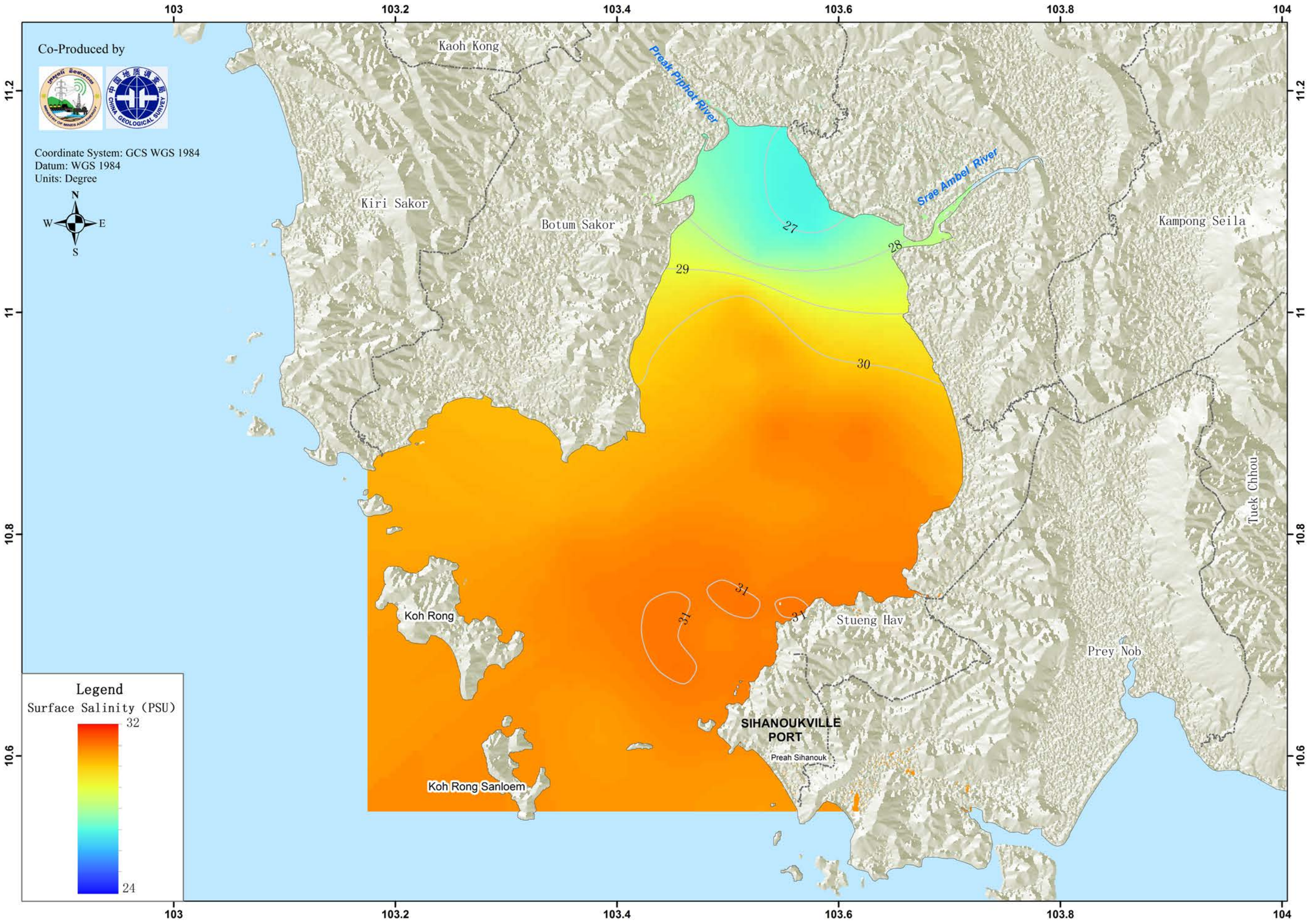
Surface Temperature



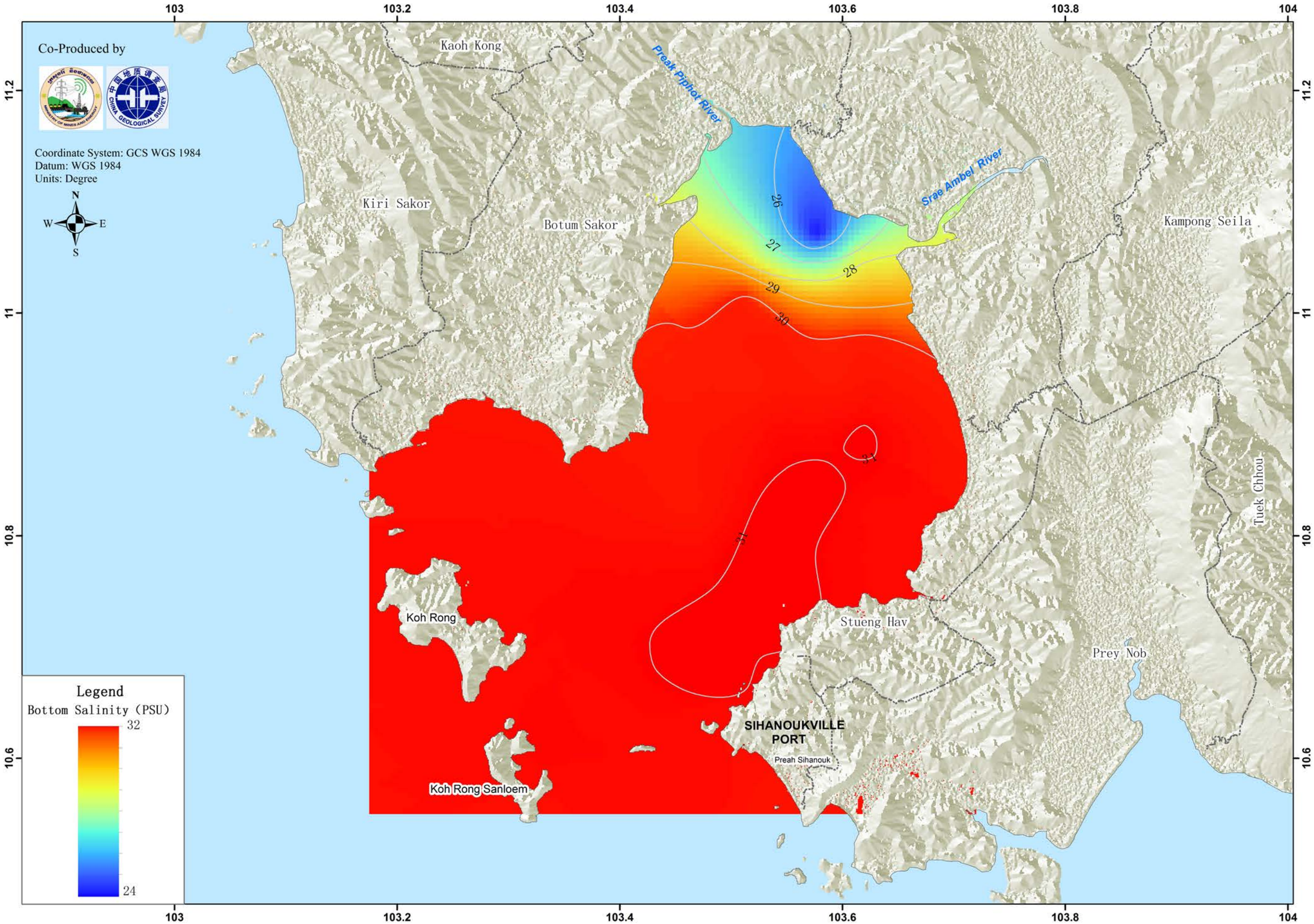
Bottom Temperature



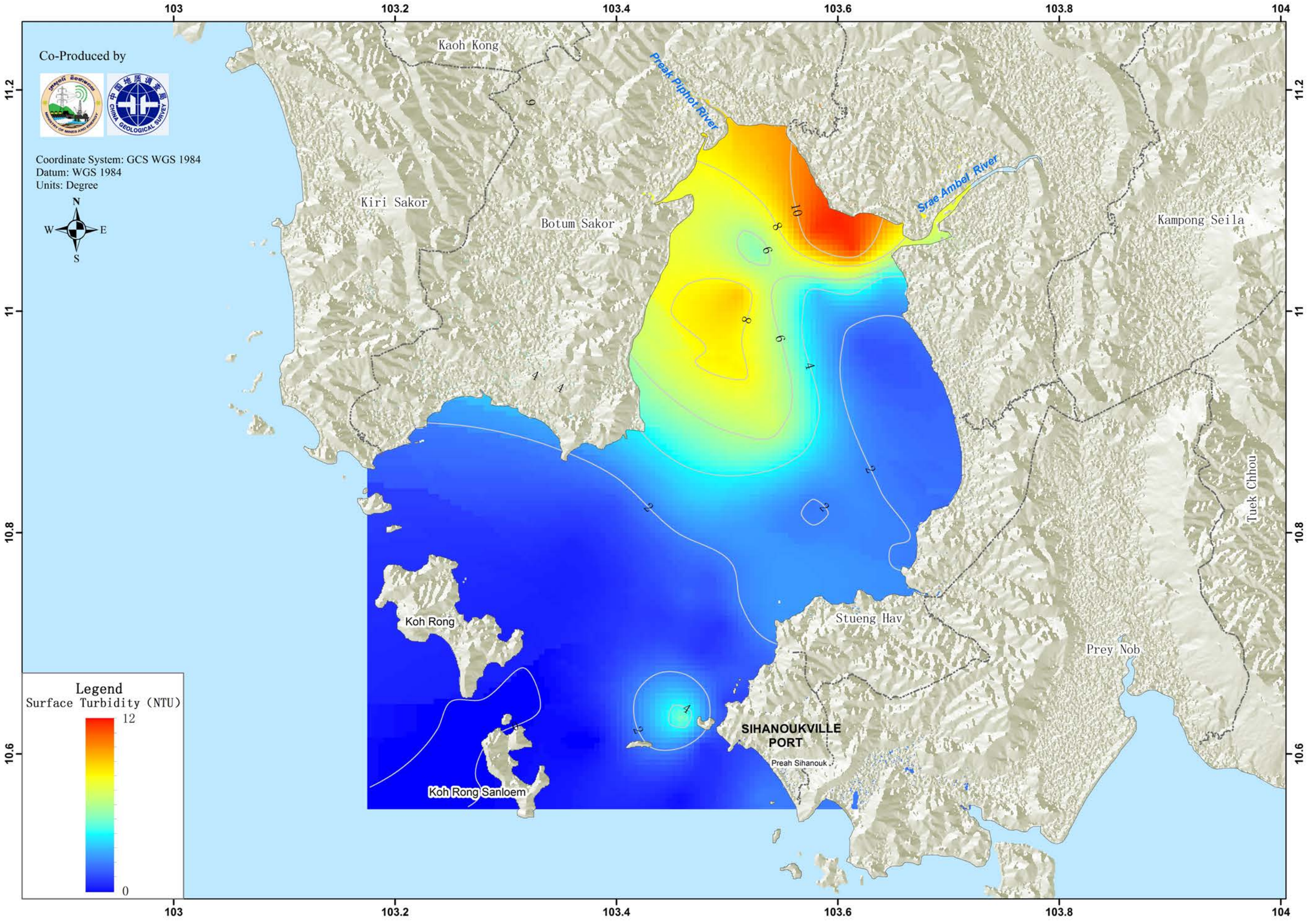
Surface Salinity



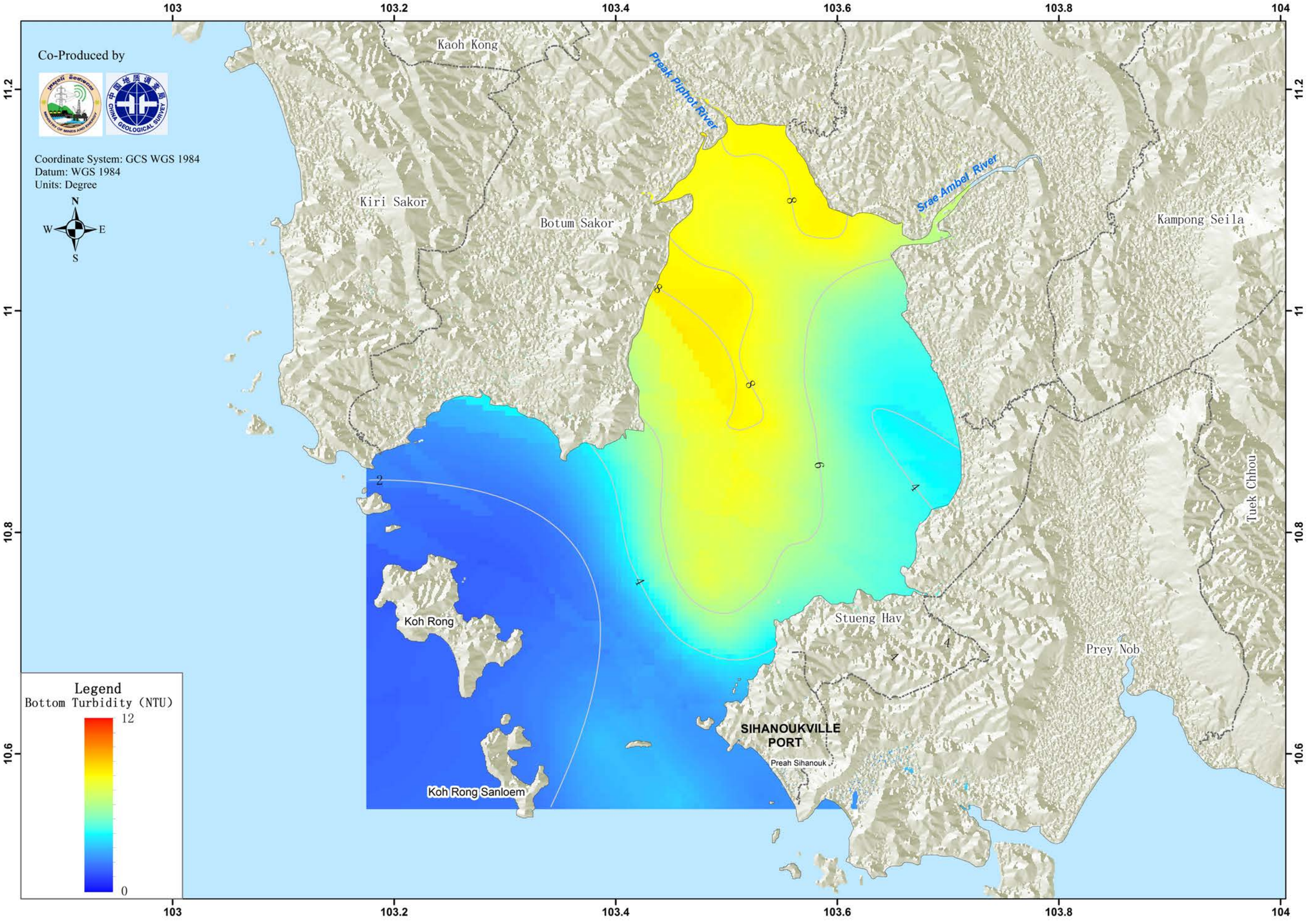
Bottom Salinity



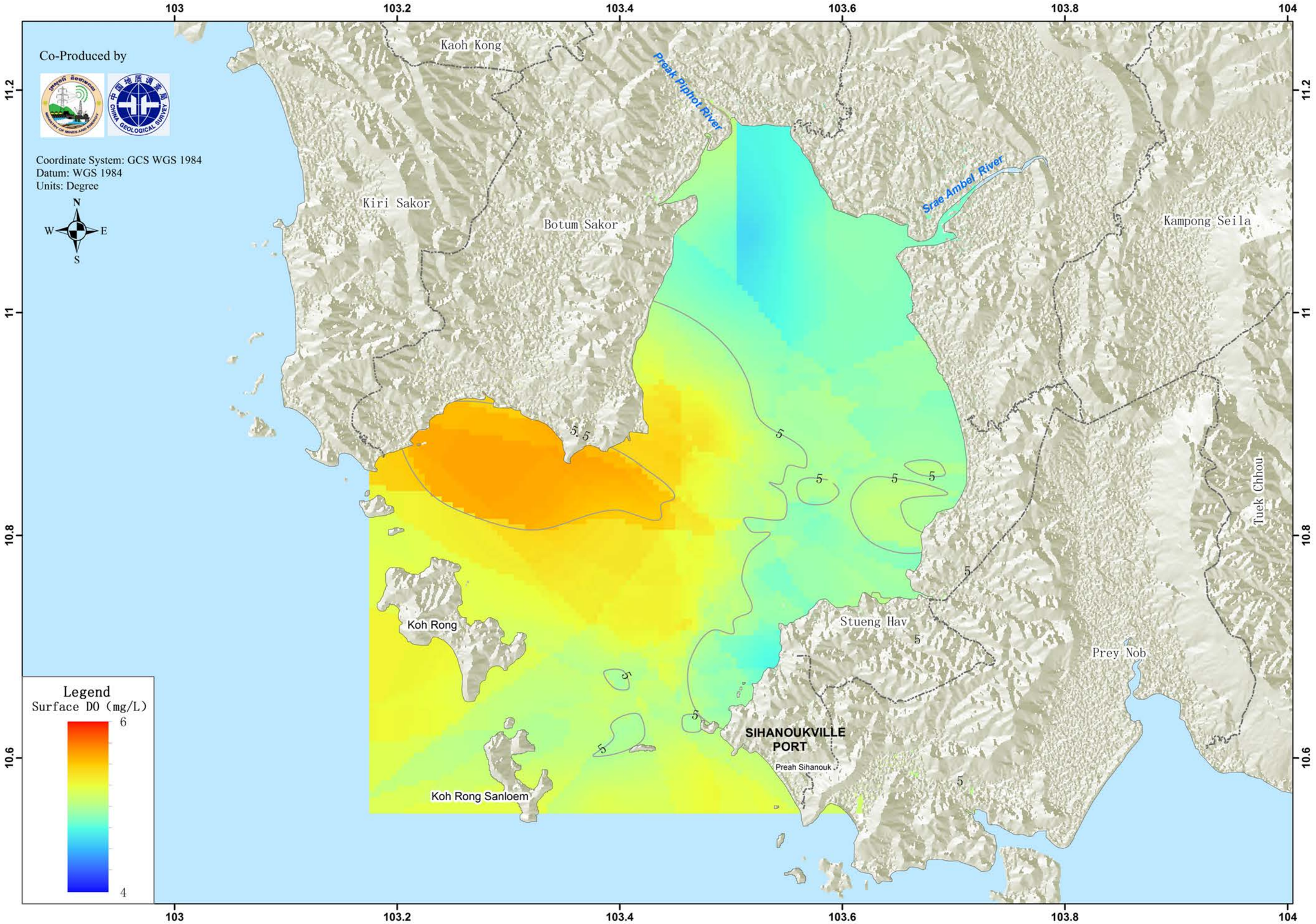
Surface Turbidity



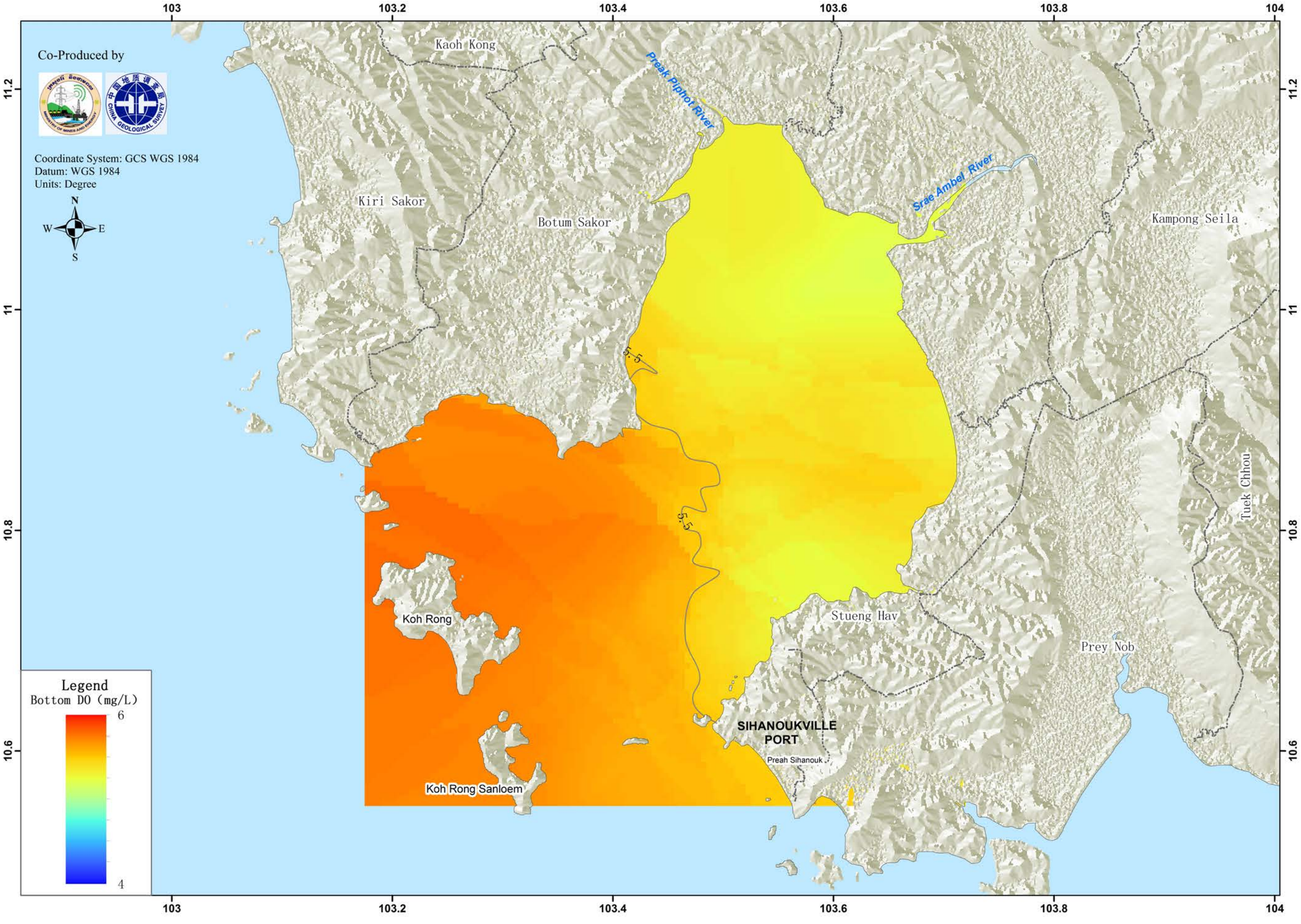
Bottom Turbidity



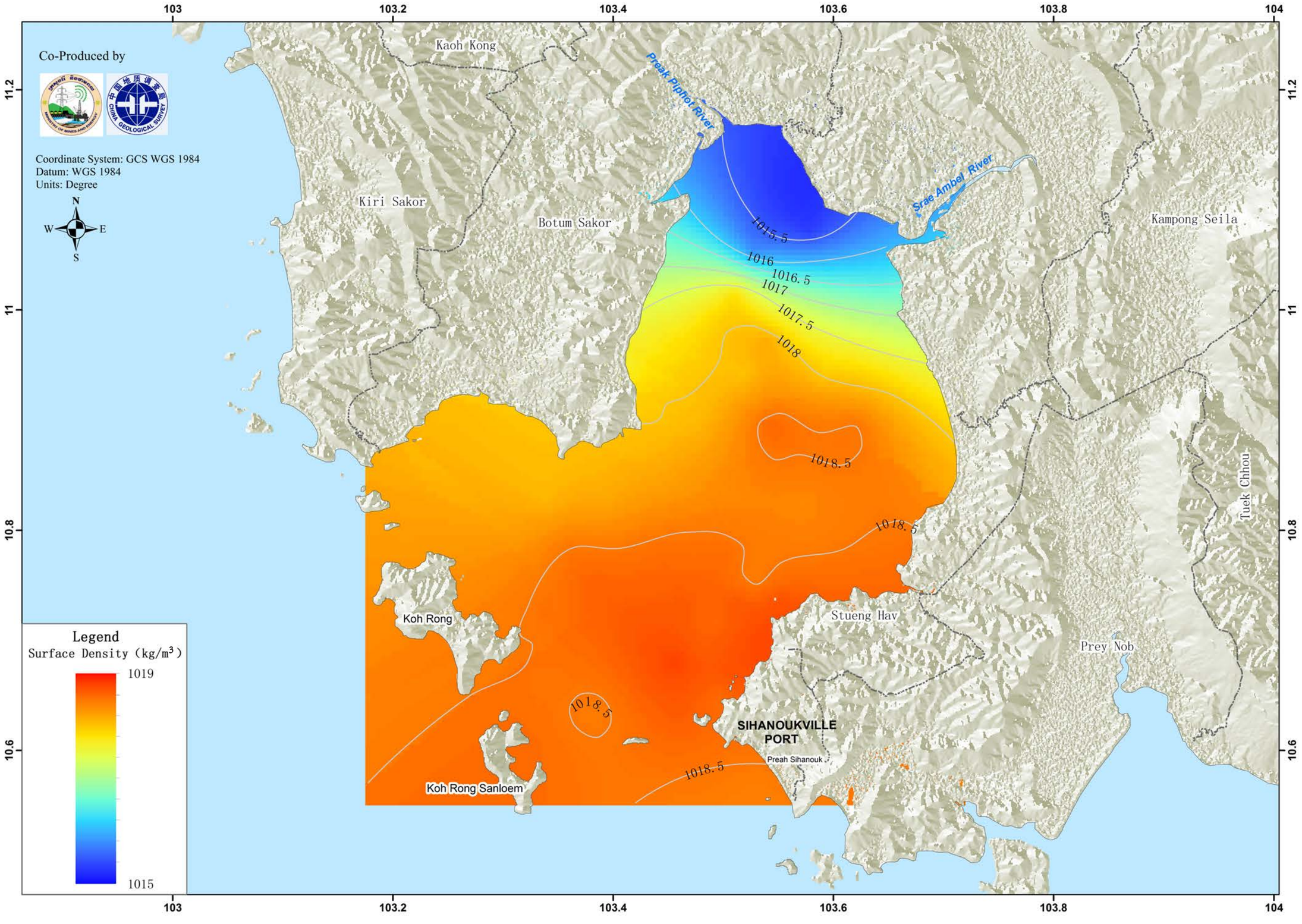
Surface Dissolved Oxygen



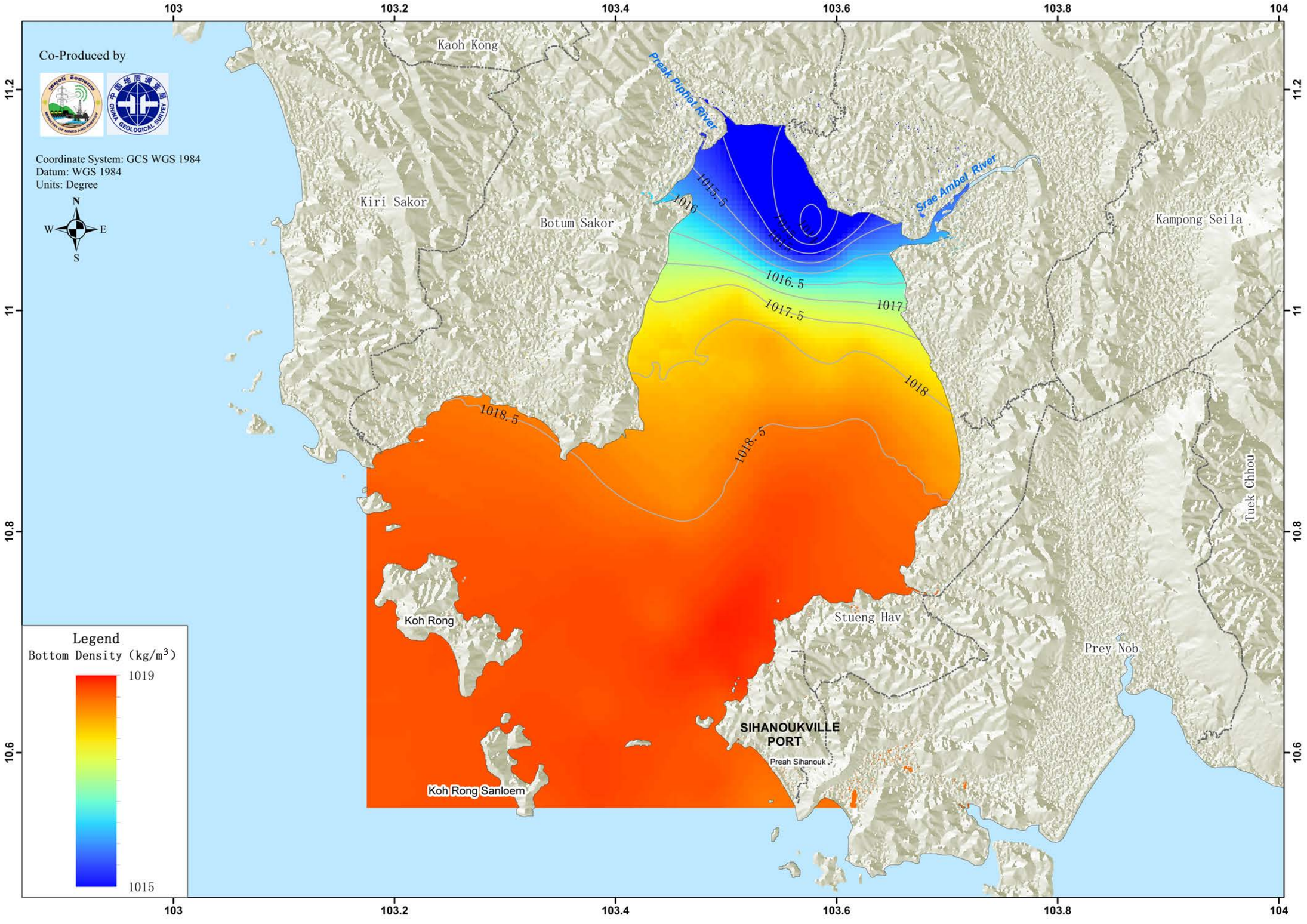
Bottom Dissolved Oxygen



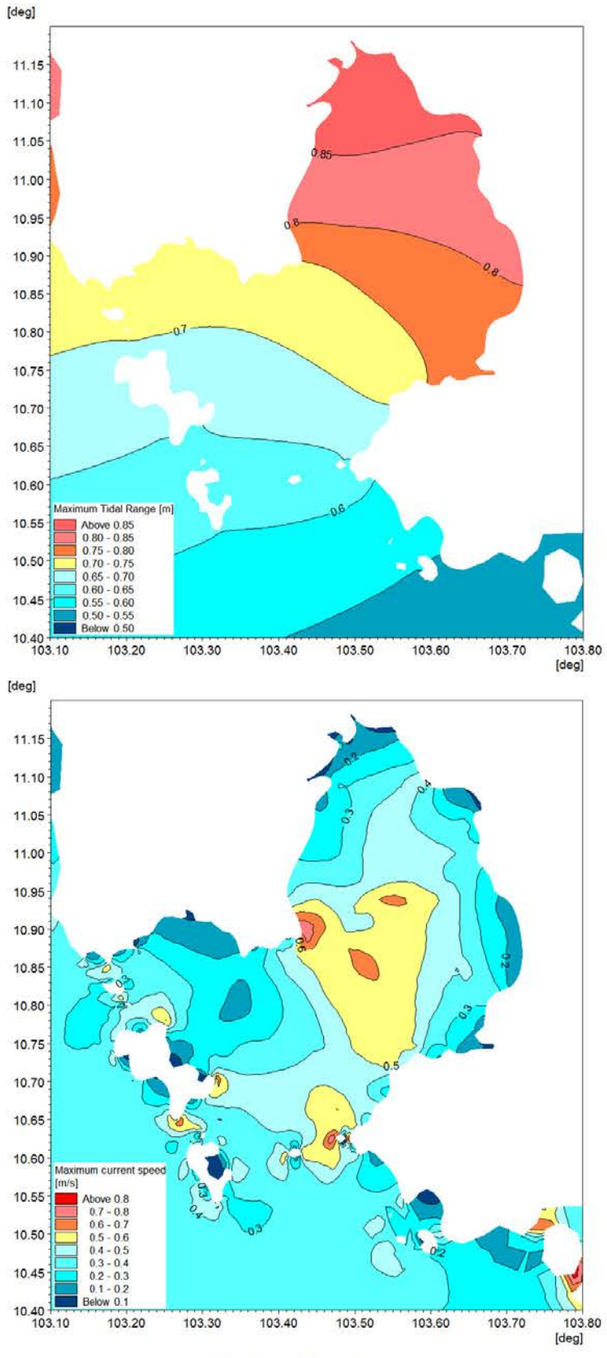
Surface Density



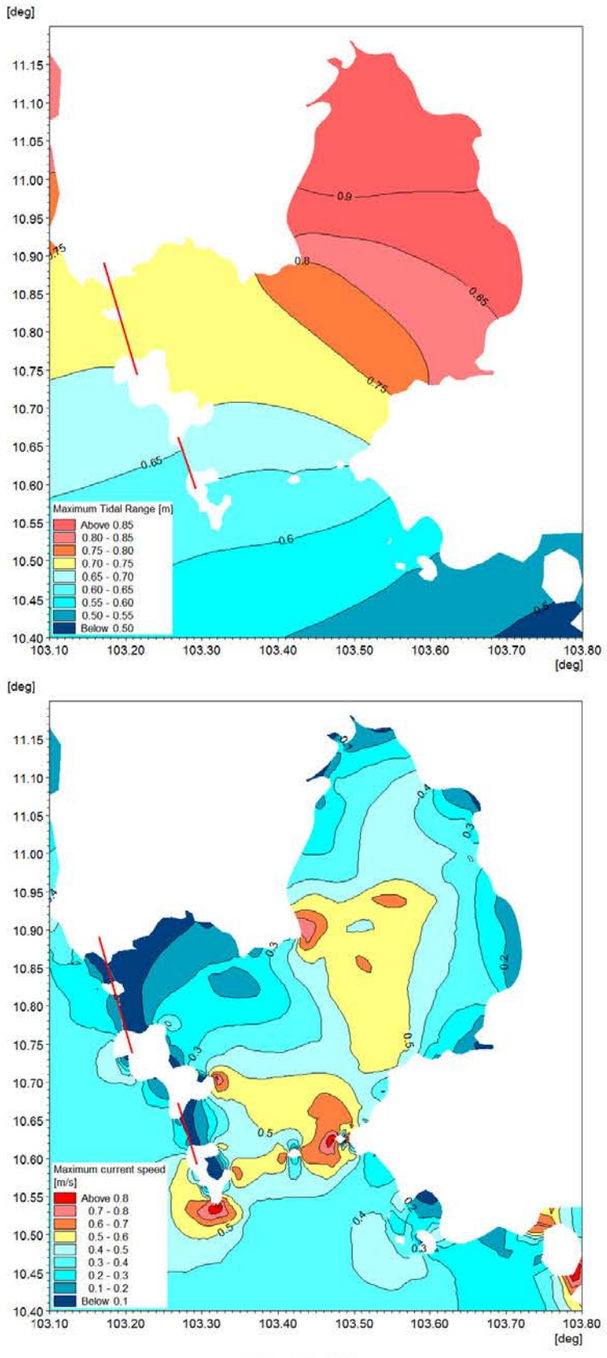
Bottom Density



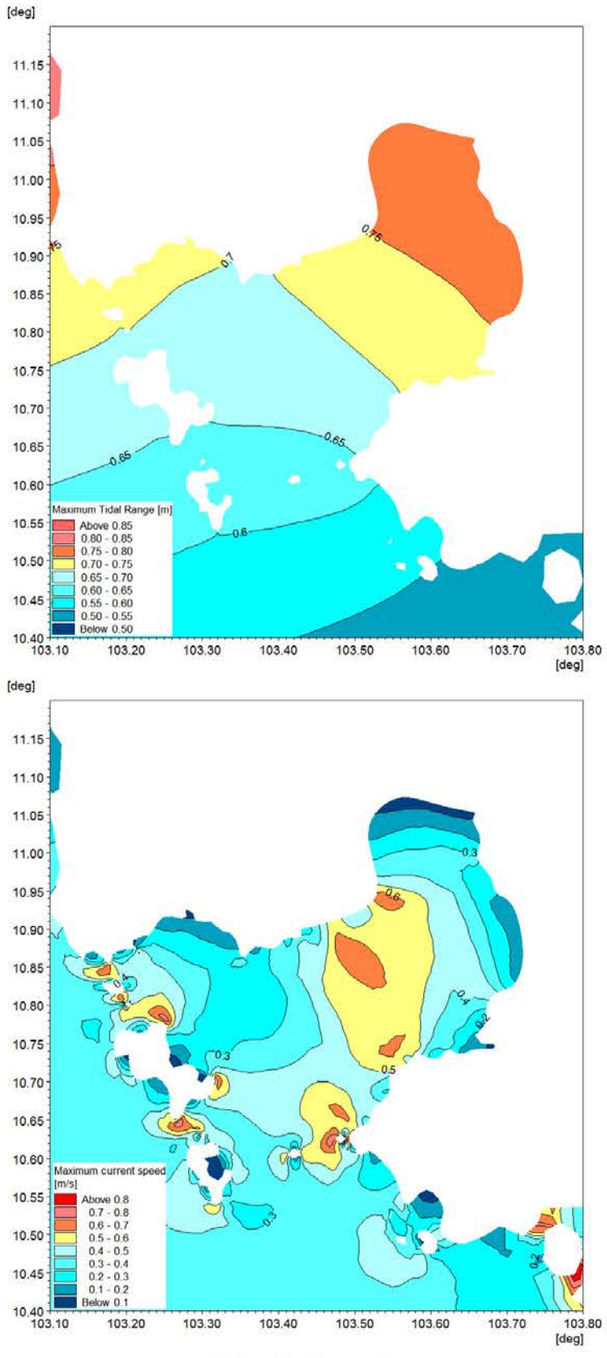
The Variation of The Tide After The Construction



Natural State

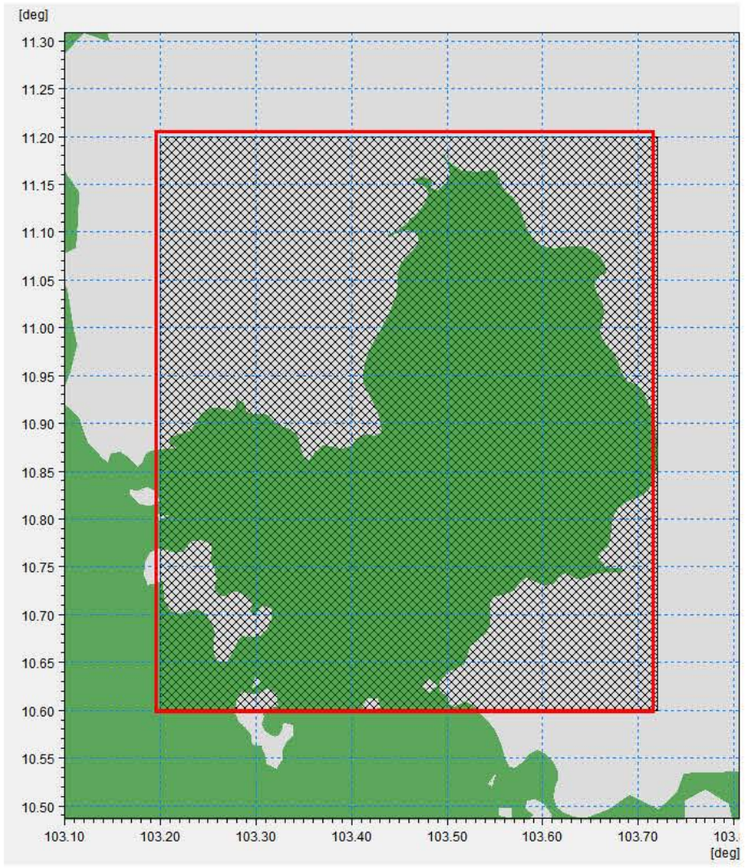


After Bridge

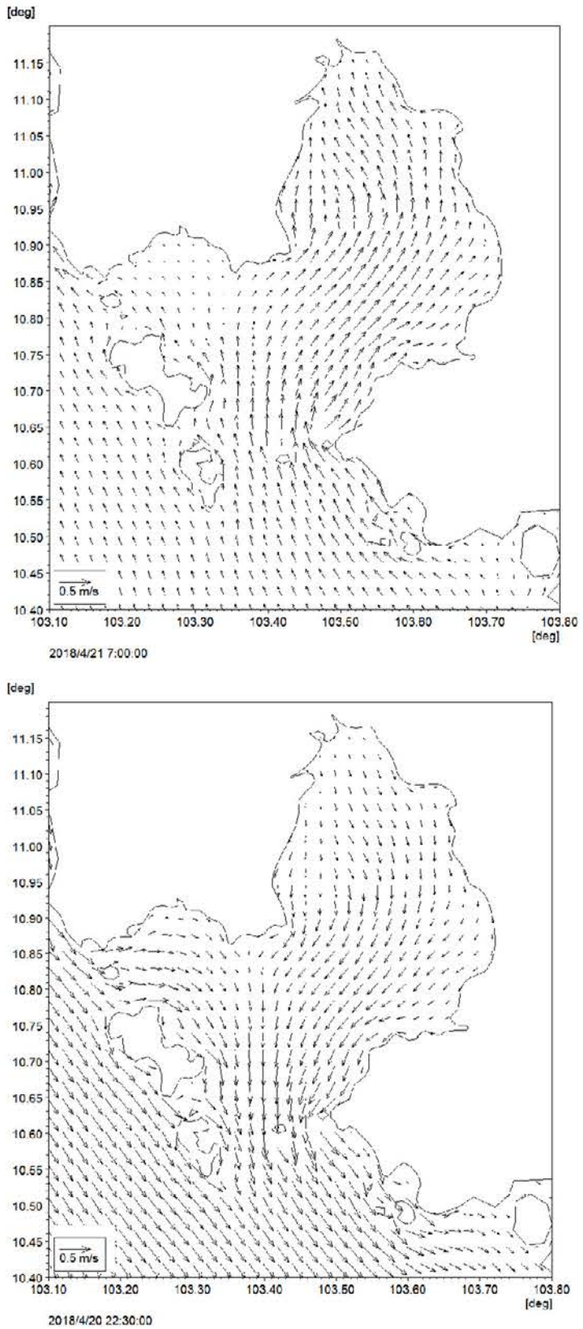


After Reclamation

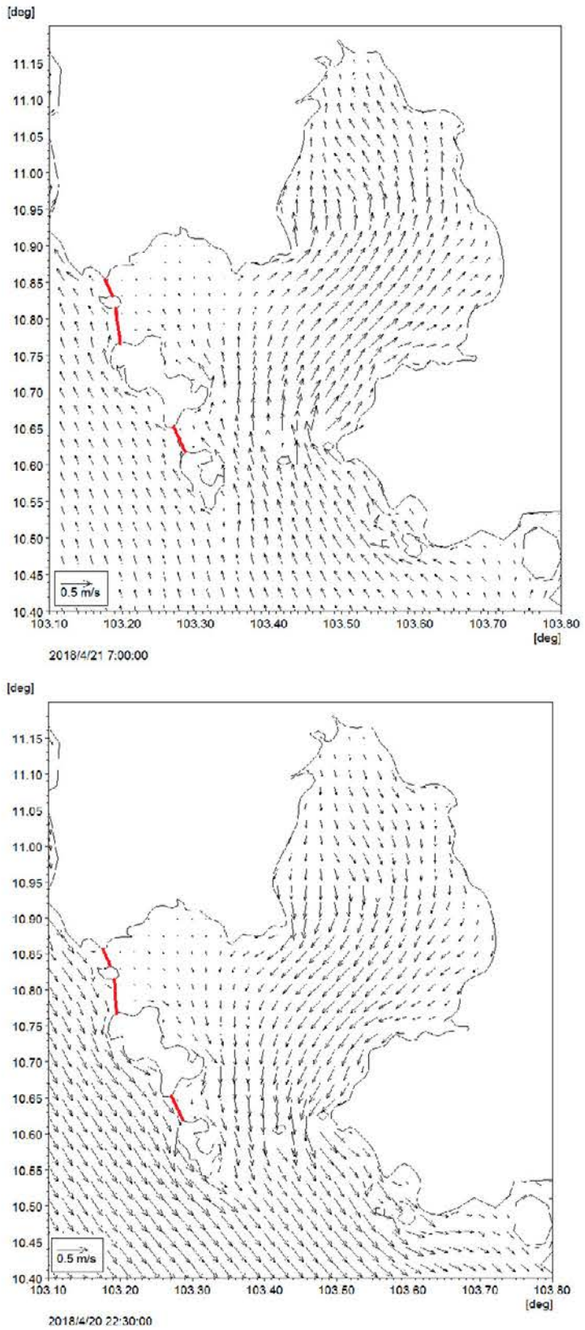
The Variation of The Tidal Prism After The Construction



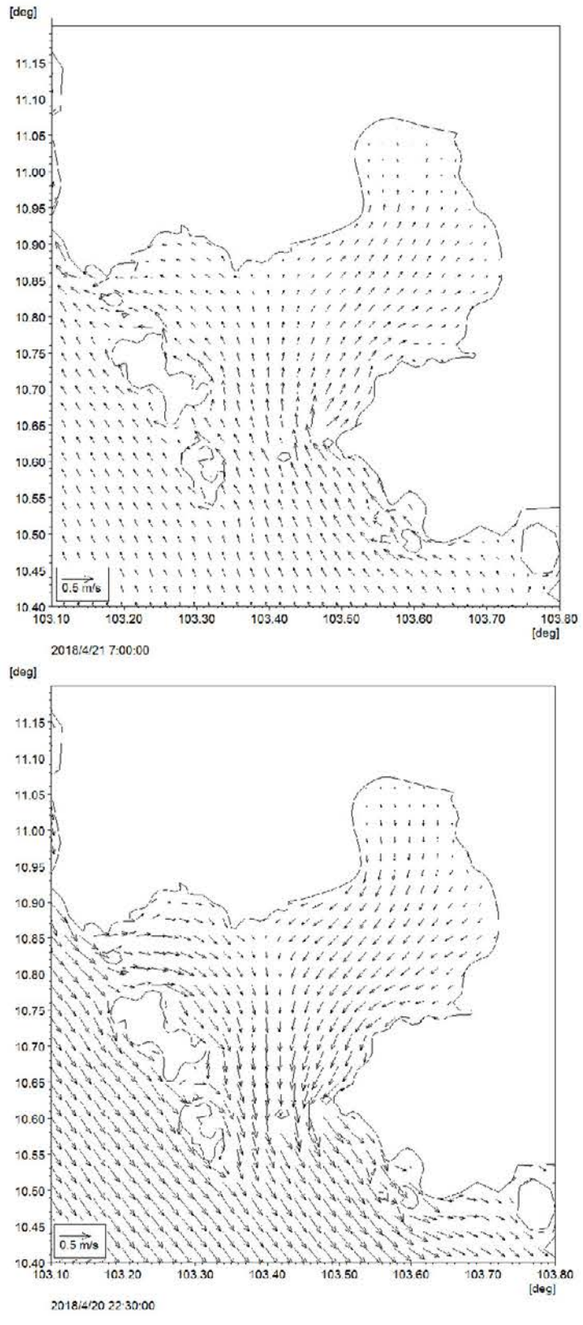
The Flow Field of The Flood and Ebb Tide



Natural State

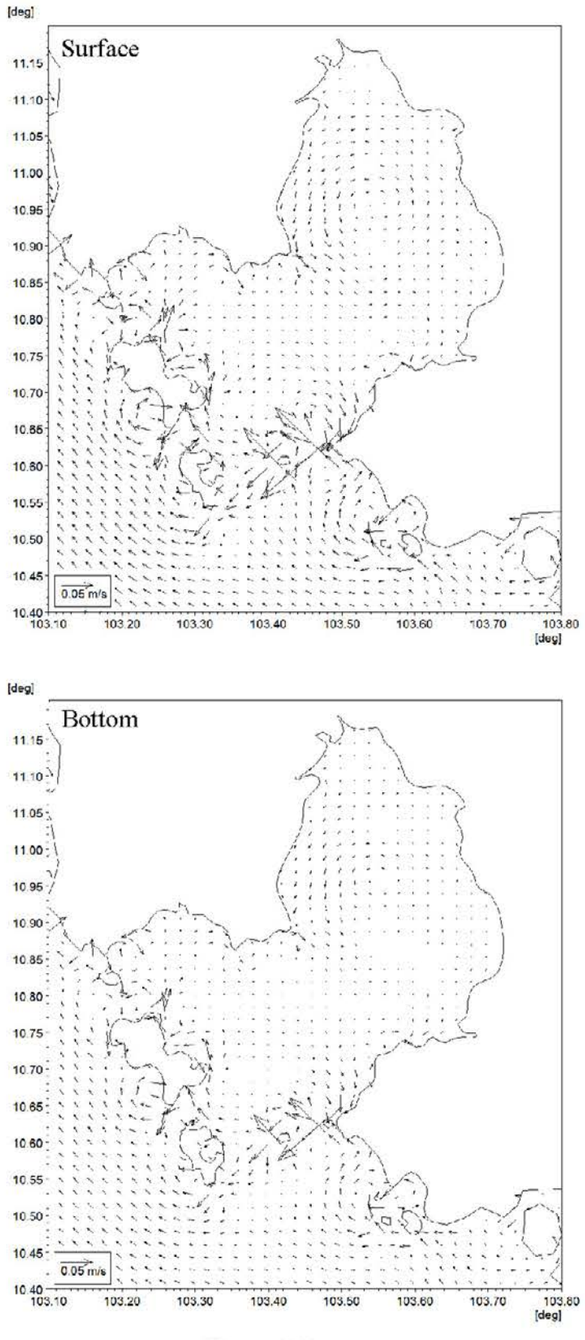


After Bridge

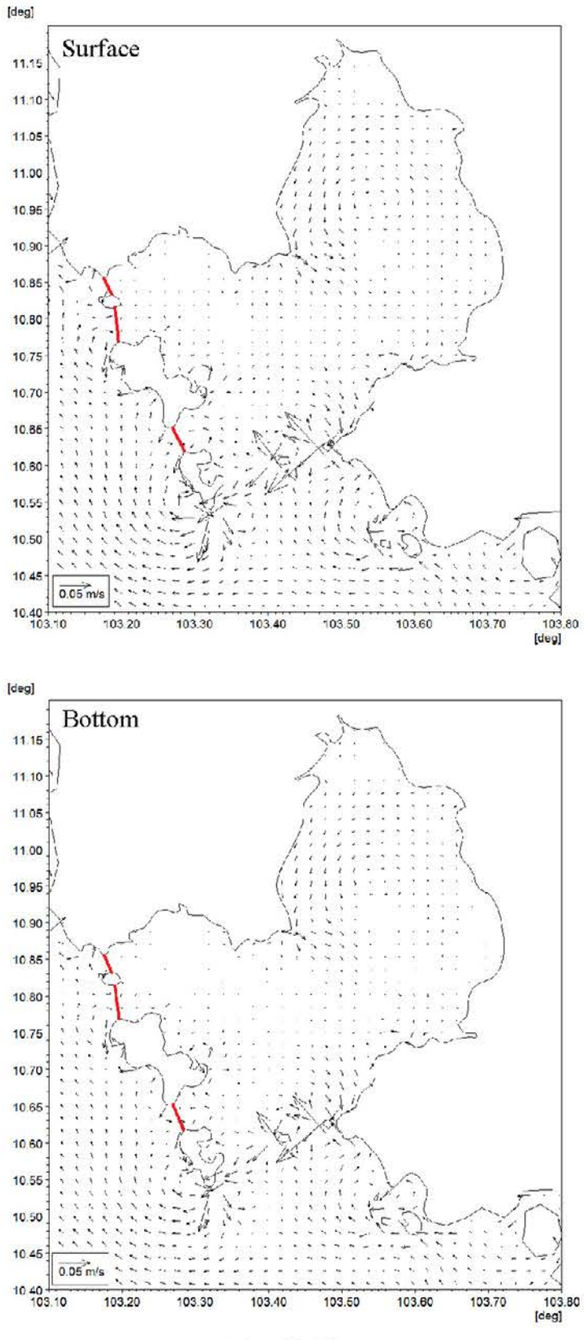


After Reclamation

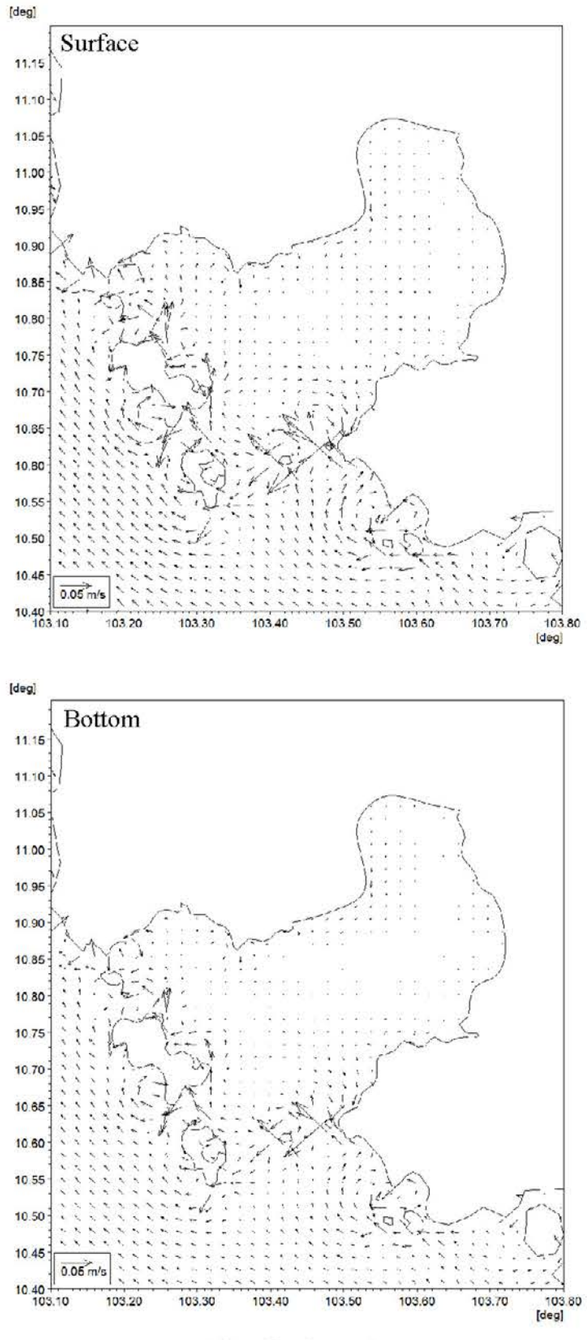
The Tidal Residual Current Field



Natural State

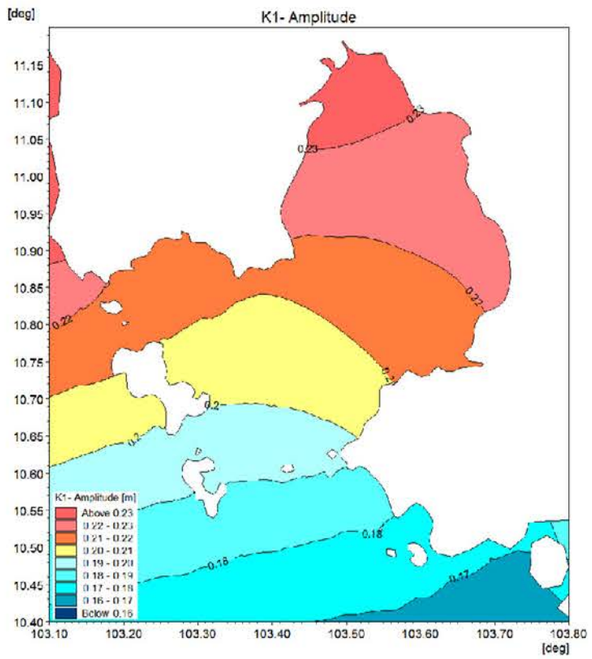


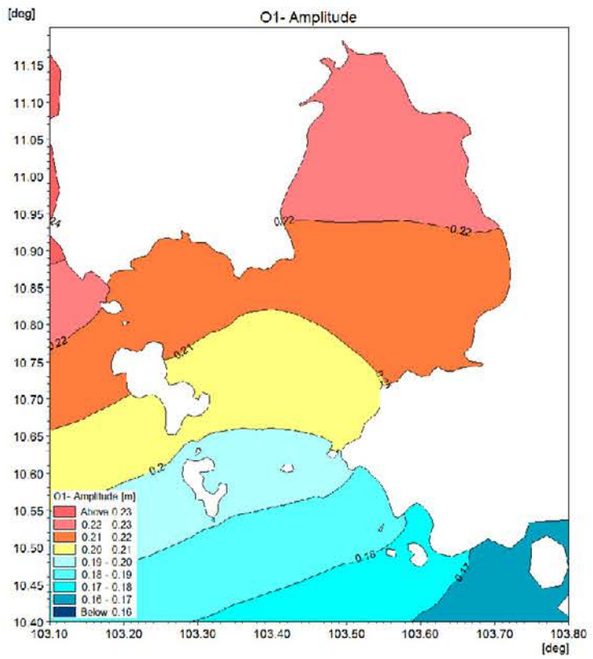
After Bridge

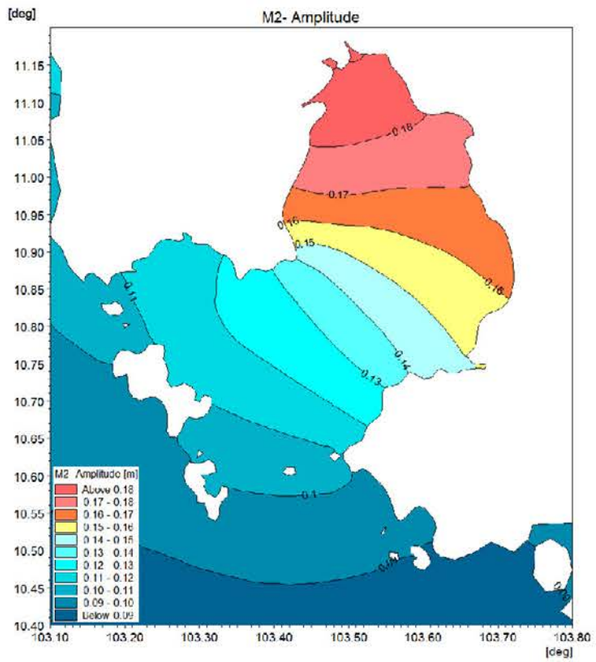


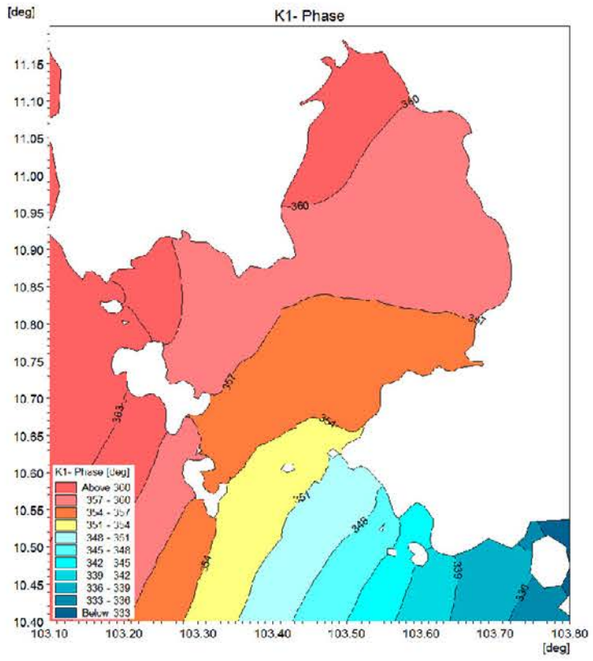
After Reclamation

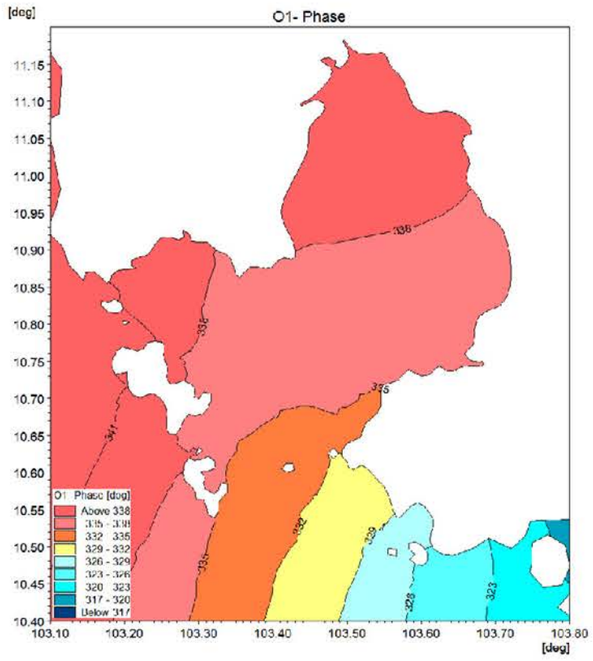
Harmonic Constant of K1、O1、M2 tide

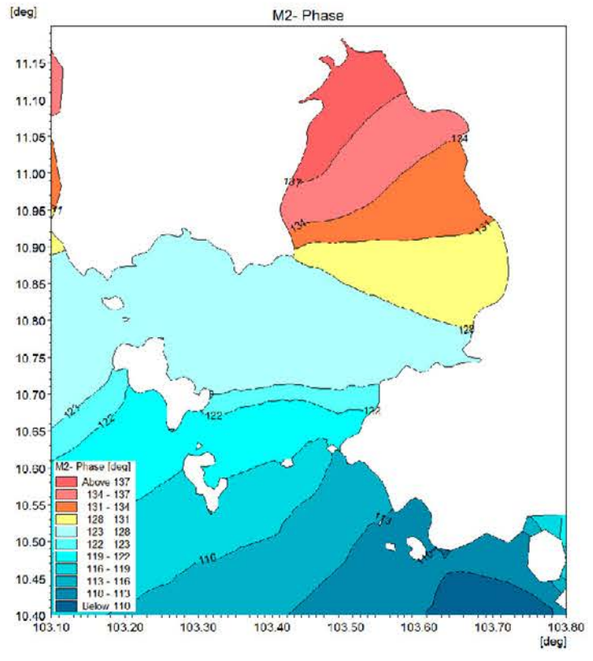




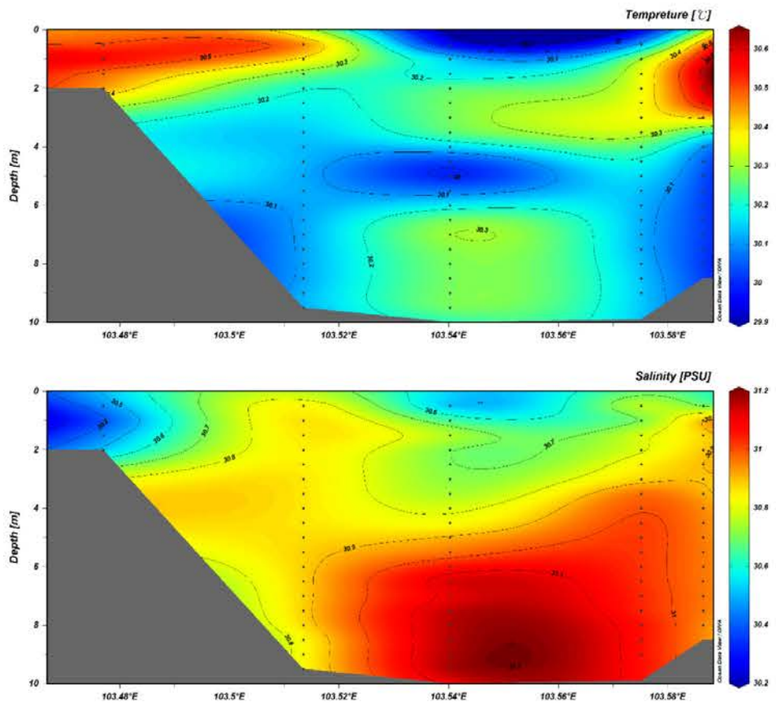




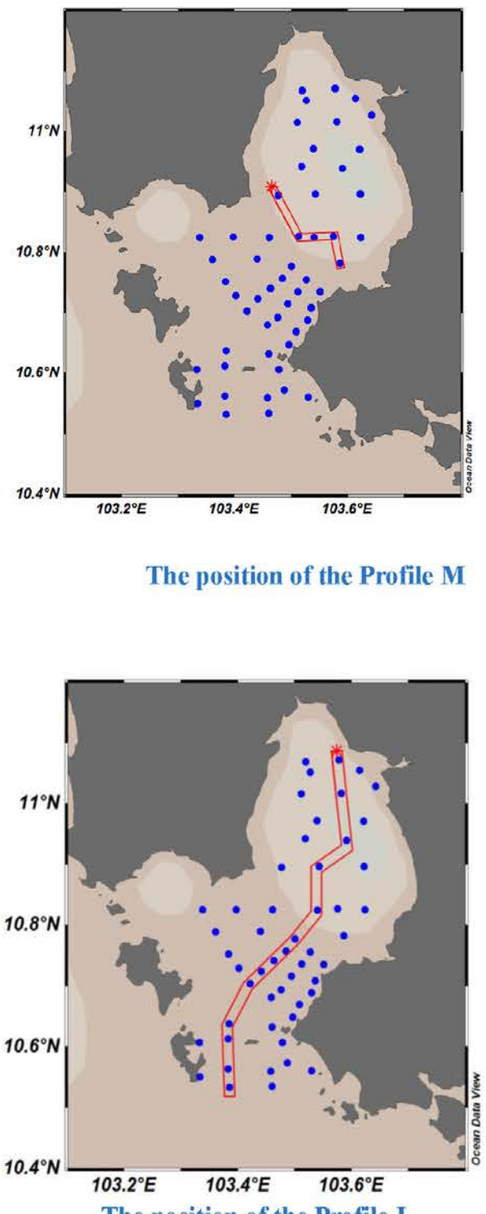




The Distribution of Hydrological Parameters Along The Profile

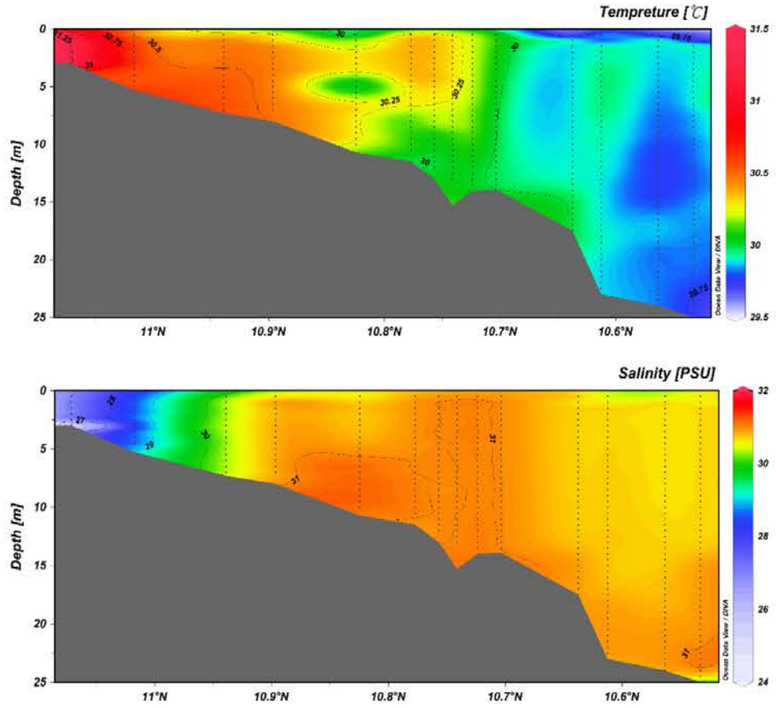


Profile M



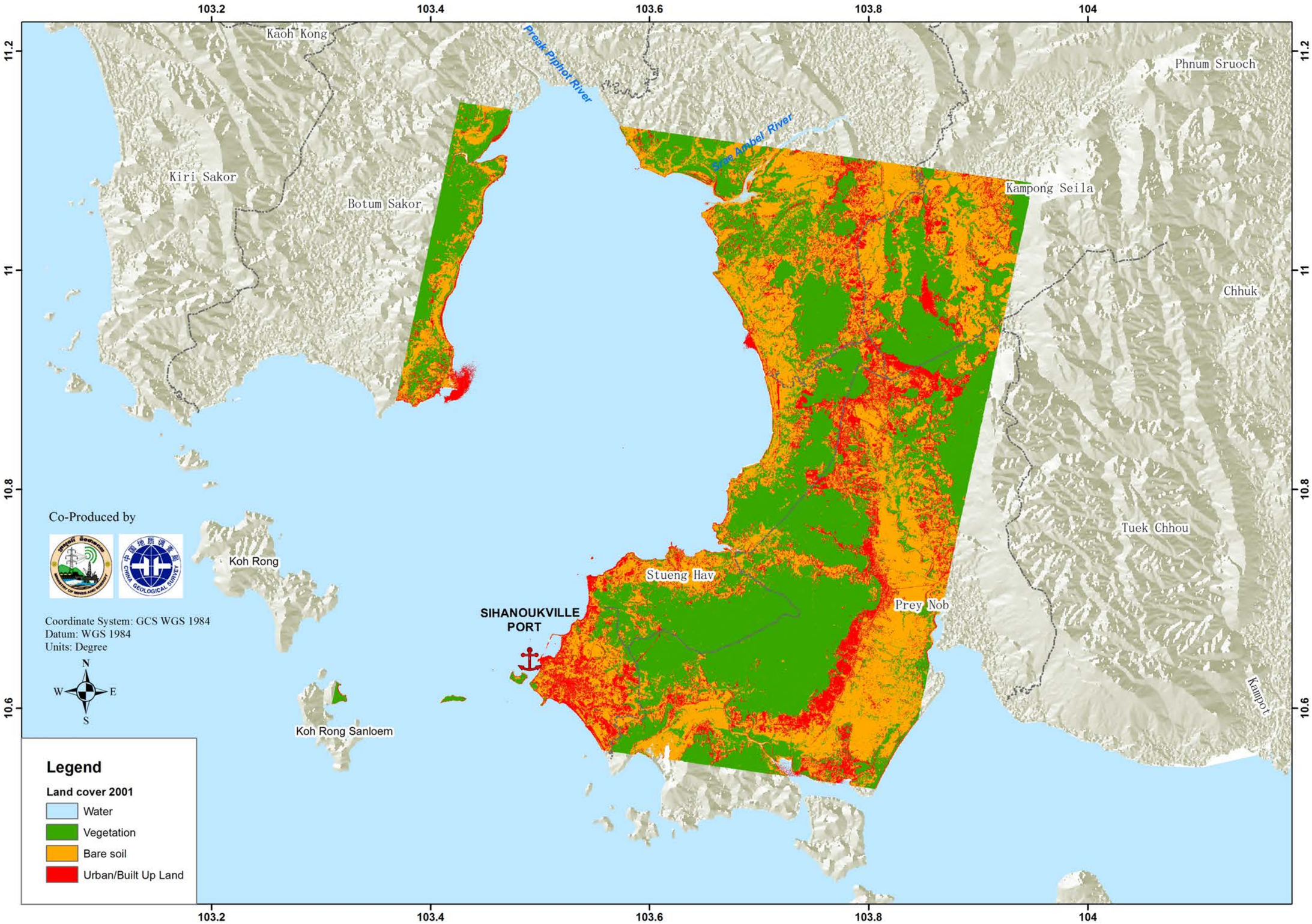
The position of the Profile M

The position of the Profile L

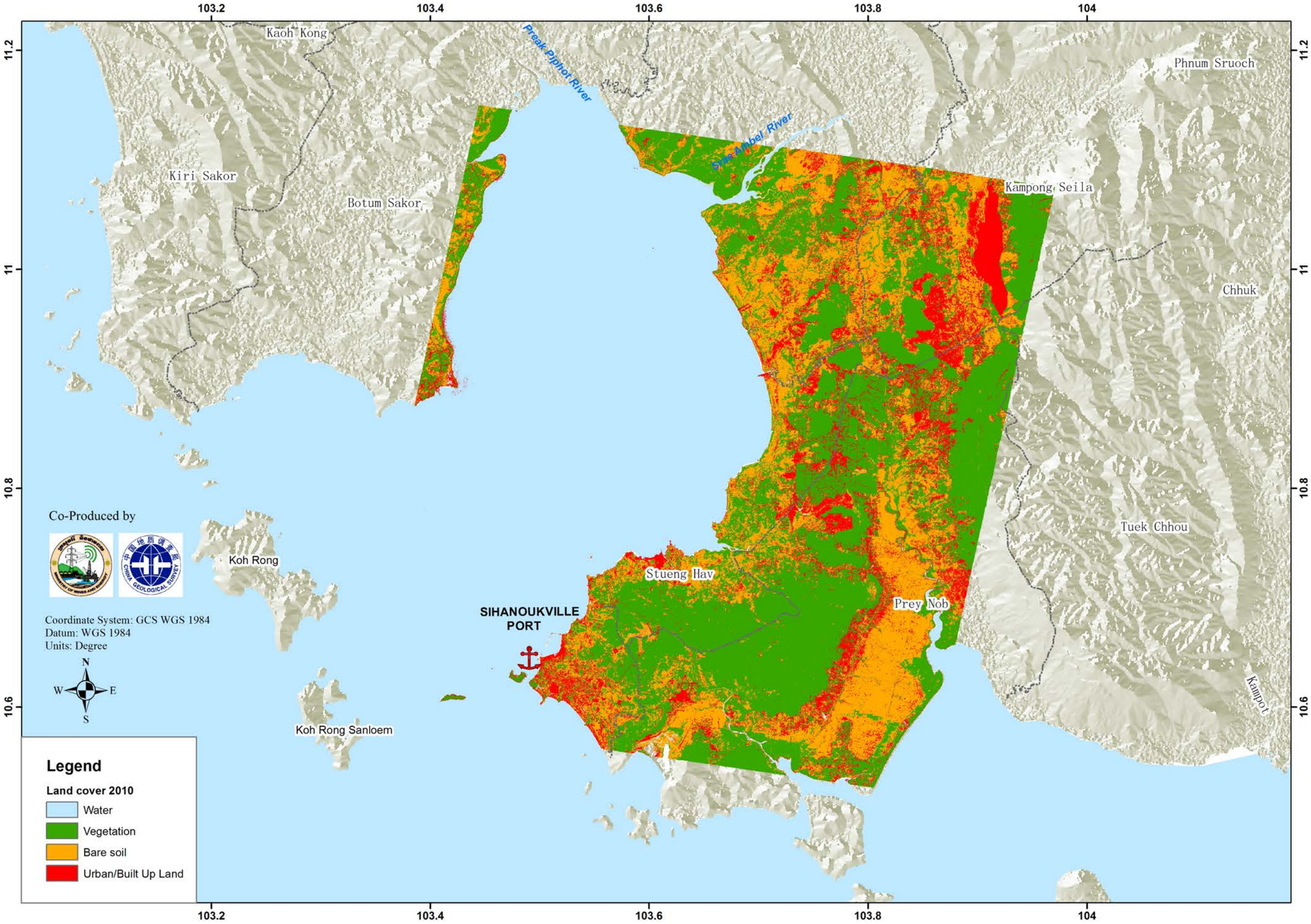


Profile L

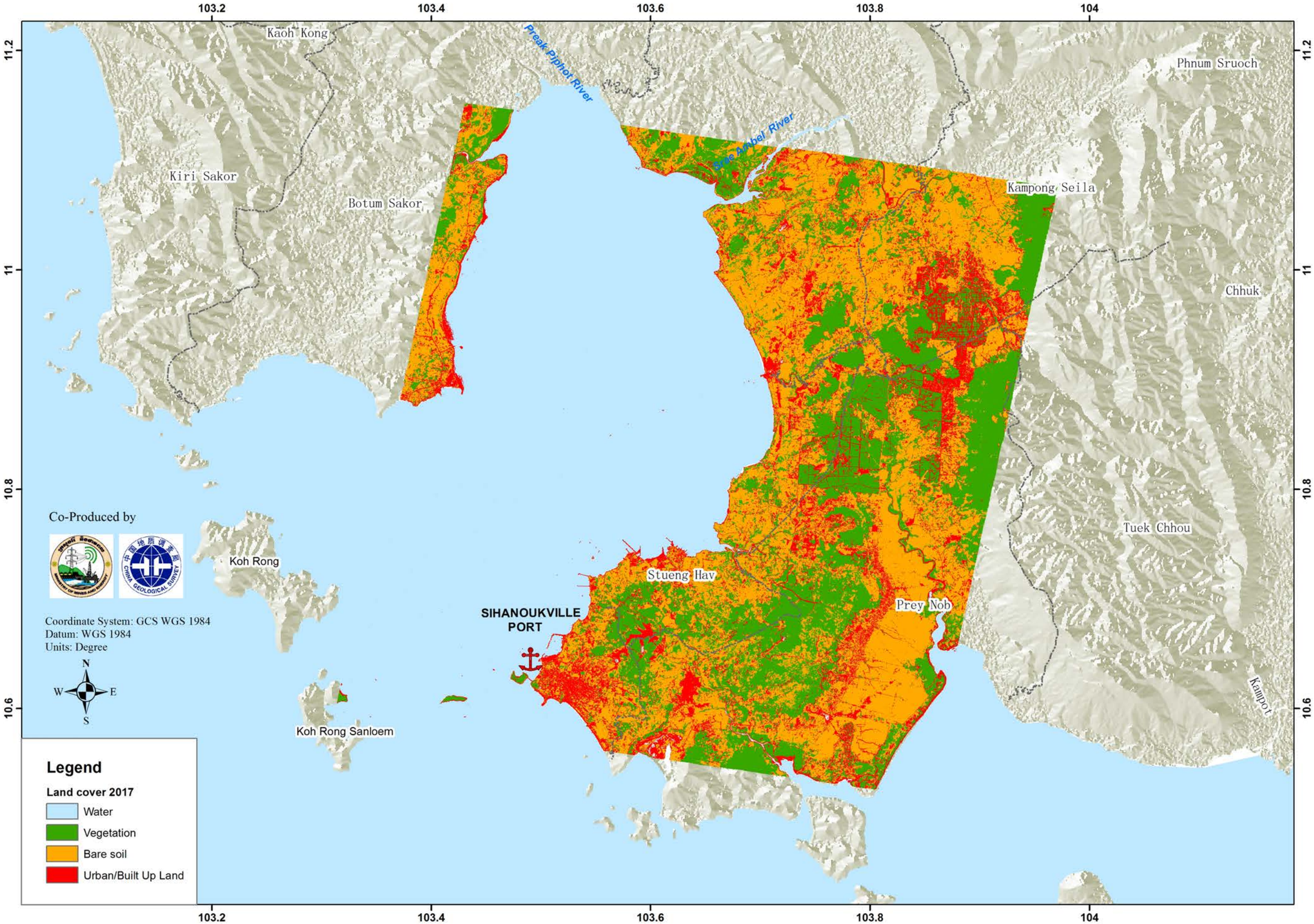
Land Utilization (2001)



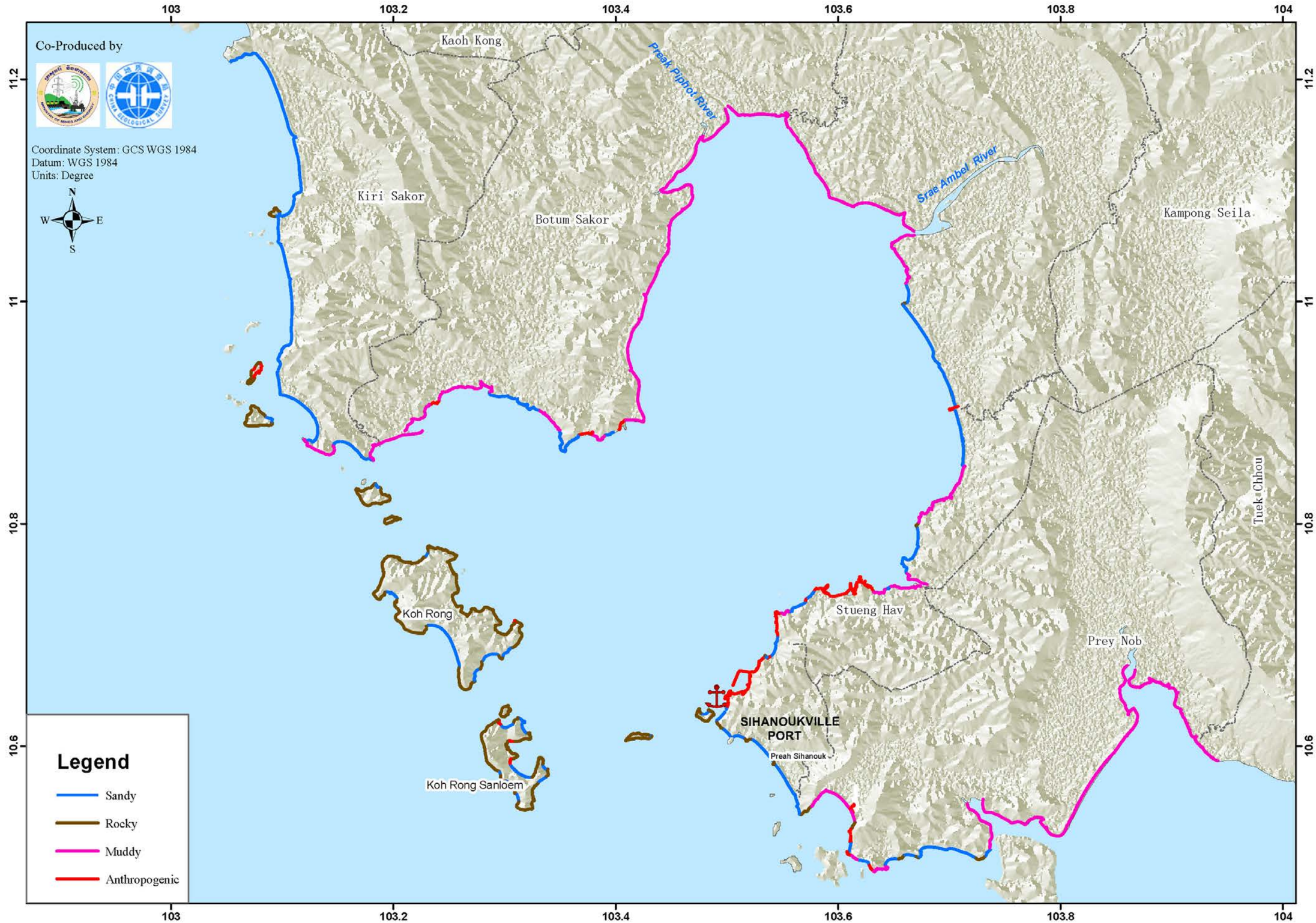
Land Utilization (2010)



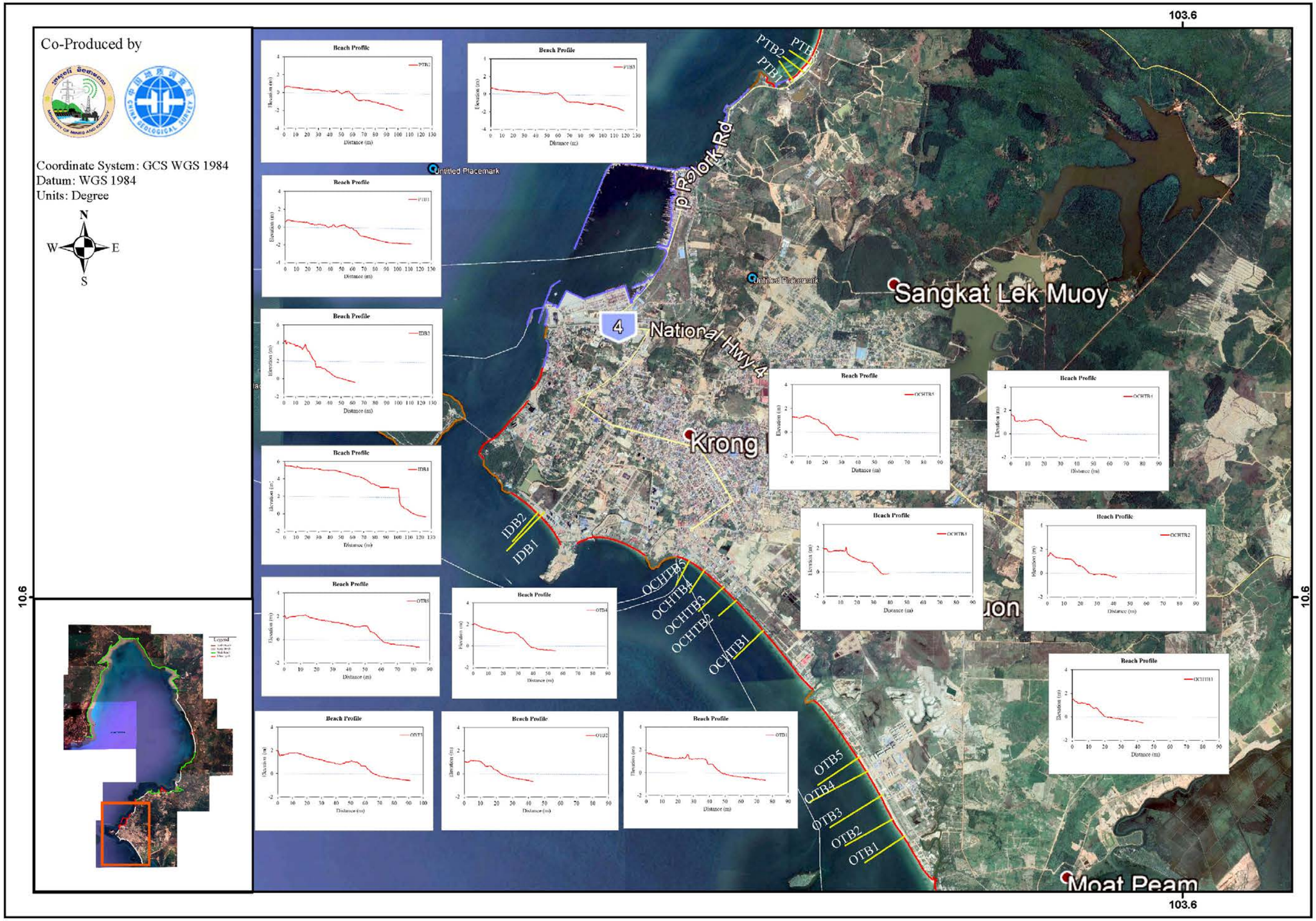
Land Utilization (2017)



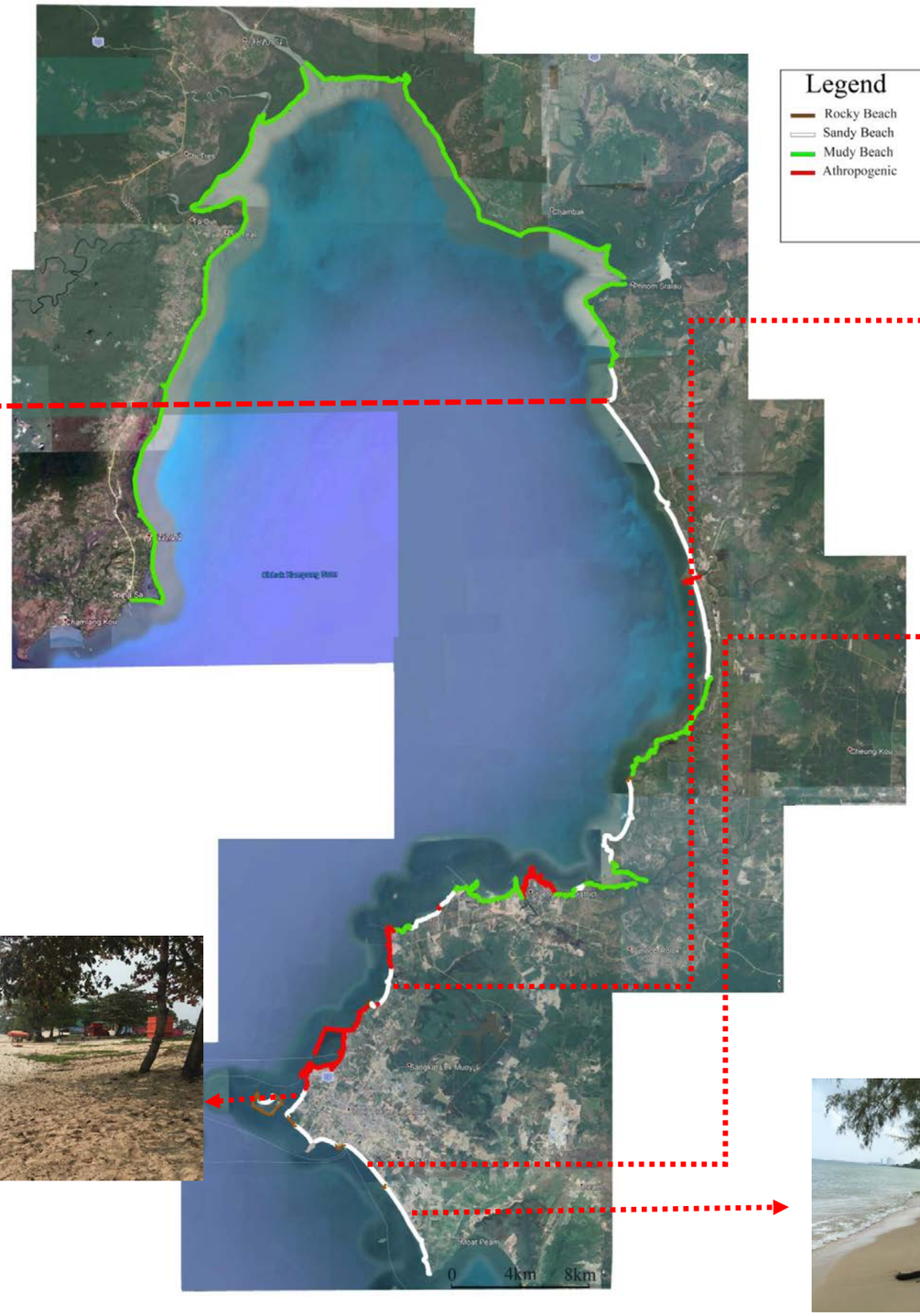
Coastal Line Classification



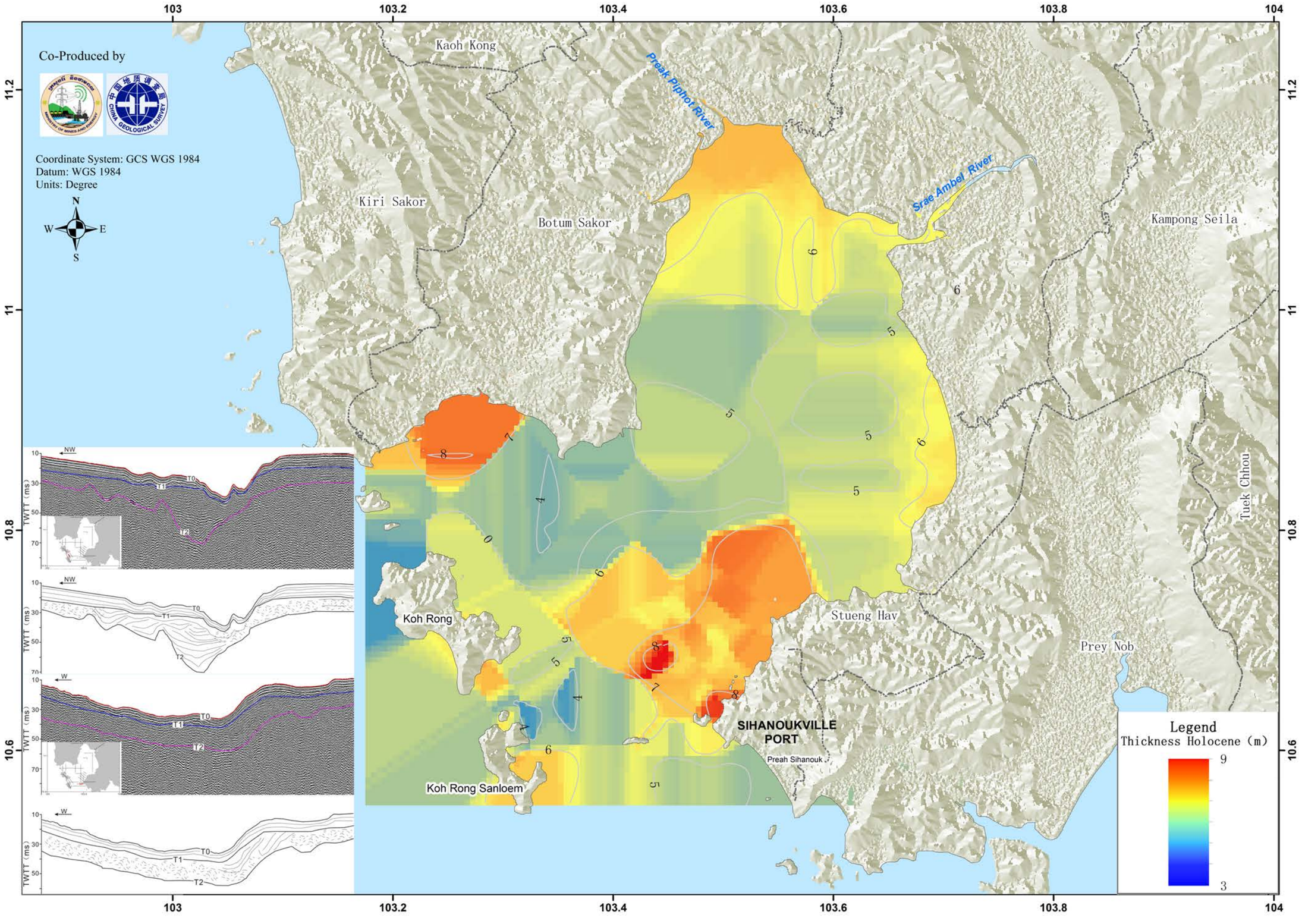
Beach Profile Map



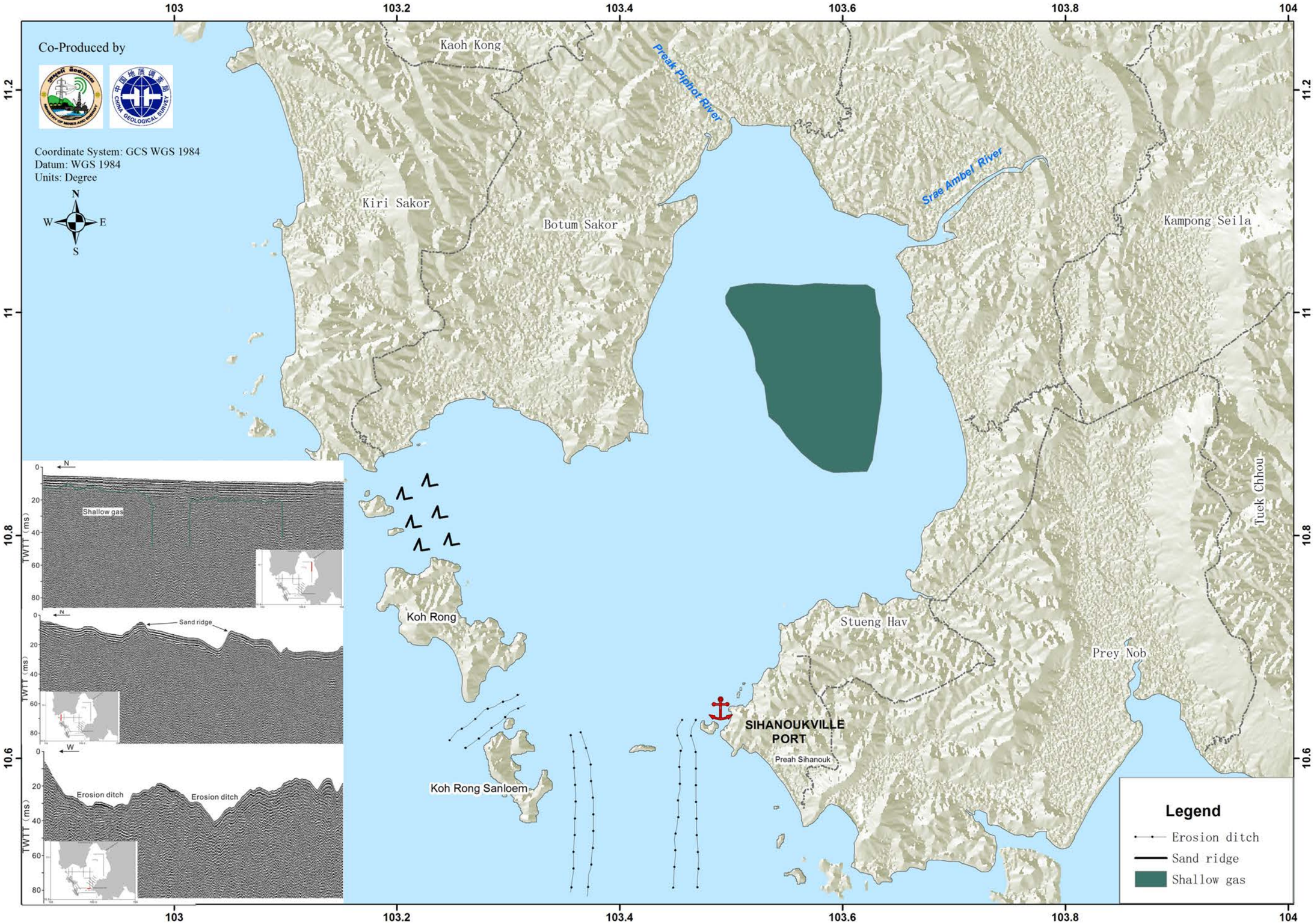
Kampong Som Bay Beach Distribution



Isopach of The Holocene Sediment



Distribution of The Geo-hazards Features



Signing Agreement and Cooperation Workshops

China-Cambodia signed Agreement on Marine Geoscience Cooperation (2017.04)

Visit General Department of Mineral Resources(2017.08)

Visit Sihanoukville Geological Survey(2017.08)

Visit Sihanoukville Port Bureau(2018.03)

Discussion of the work proposal with GDMR(2018.03)

Discussion of the work proposal with Sihanoukville Geological Survey (2017.08)

Visit Sihanoukville Port Bureau(2018.03)

Visit Sihanoukville Port Bureau(2018.03)

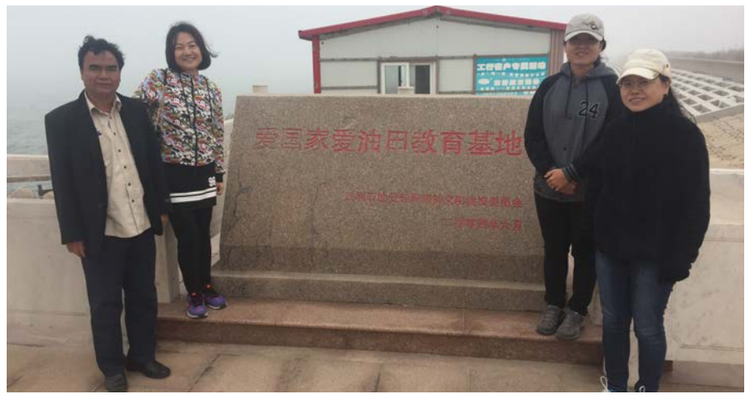
Discussion of the work proposal with Sihanoukville Geological Survey (2018.03)

China-Cambodia Joint Field Investigation

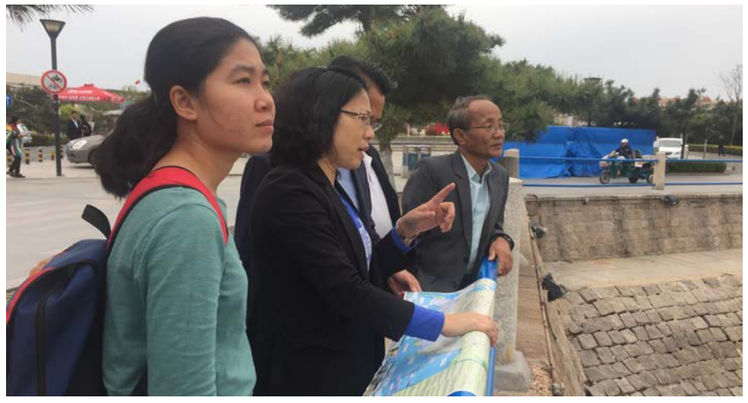
Investigation of coastal erosion at Qingdao(2017.4)

Investigation of coastal utilization at Qingdao(2017.4)

Investigation of saltwater intrusion at Laizhou Bay(2017.4)



Investigation of coastal protection engineering along the Yellow River Delta(2017.4)

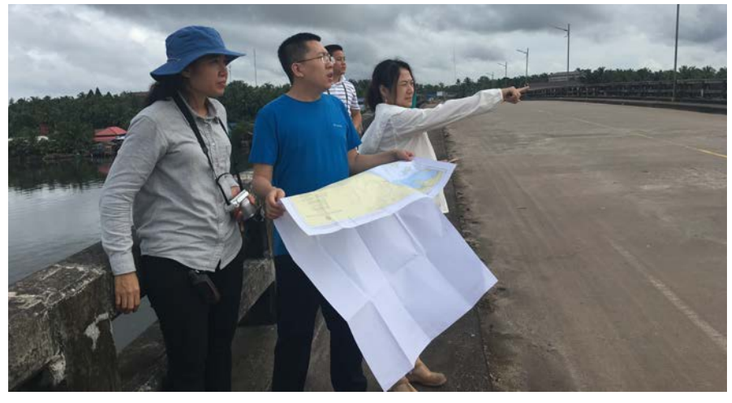


Investigation of coastal utilization at Qingdao(2017.4)

Investigation of wetland in the Yellow River Delta(2017.4)

Investigation of coastal utilization at Kampong Som Bay(2017.8)

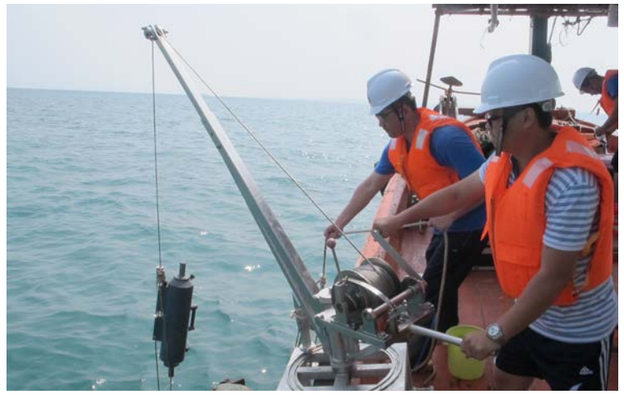
Investigation of Srae Ambel River Estuary at Kampong Som Bay (2017.8)



Investigation of the Preak Piphot River Estuary at Kampong Som Bay (2017.8)

China-Cambodia Joint Coastal Geological Survey at Kampong Som Bay,Cambodia(2018.04)

Marine geological survey at Kampong Som Bay(2018.04)

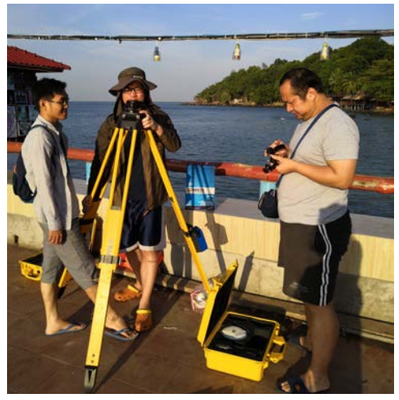


Sedimentary dynamics survey

Sampling of surface sediments on the seafloor

Survey equiments arriving in Sihanoukville, Cambodia

Marine geological survey at Kampong Som Bay



Beach monitoring

Shallow seismic profile survey

Shallow seismic profile survey

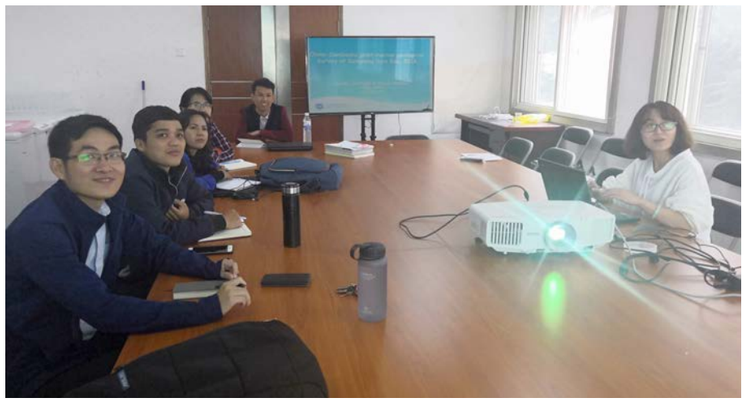
Beach sediment sampling

Technical Training and Personnel Exchanges(2018.10-2018.12)

Training of beach monitoring data analysis

Data analysis training of sediment dynamics

Training of grain size of sediment data processing



Training of remote sensing images data processing

Collaborative research

Training of the data analysis of shallow seismic profile survey

Training of samples process methods in laboratory

Discussion on the preparation of result atlas

Training of drilling cores description and sediment samples selection